

Research papers

Modelling heat conduction in 3D composite cathode microstructures of all-solid-state batteries

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ABSTRACT

All-solid-state lithium batteries (ASSLBs) are a promising next generation energy storage technology comparing to conventional lithium-ion batteries (LIBs). Although ASSLBs have high thermal stability, thermal degradation and thermal runaway can still occur. The thermal characteristics of the cathode of ASSLBs play a crucial role in maintaining the stability of the interface with the electrolyte. It is important to understand the thermal characteristics of ASSLBs, which is highly associated with specific microstructure geometries of composite cathodes. Here, this paper presents a 3D lattice Boltzmann heat conduction model to simulate the effective thermal conductivity (ETC) of the multiphase solid-state cathodes, which is composed of active material LCO (LiCoO_2) and solid electrolyte LLZO ($\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$), generated using the discrete element method (DEM) with different porosities, volumetric ratios, particle size ratios, and various composite tortuosities. The findings indicate that porosity, volumetric fraction, and particle size all exert the decisive factor on ETC. Tortuosity emerges as a non-negligible factor influencing thermal conductivity, highlighting the importance of microstructural optimization.

1. Introduction

Rechargeable batteries with high safety and long lifetime are urgently needed for widespread deployment of electric vehicles and renewable grid storage [1]. In the last two decades, conventional lithium-ion batteries (LIBs) with liquid electrolyte dominate the market. However, the significant battery performance degradation and severe safety concerns triggered by thermal runaway (TR) are still the major challenges [2,3], mostly due to the thermal and electrochemical instability and high flammability of liquid electrolyte. All-solid-state lithium batteries (ASSLBs) could be the potential solution to overcome safety hazards associated with liquid electrolyte, as solid electrolytes (SE) present advantages of lower flammability, wider electrochemical stability window and better mechanical integrity to resist dendrite over a wide temperature range (from -50°C up to 1000°C) [4]. Despite these advantages, ASSLBs are not immune to thermal issues. Thermal degradation and localized hot spots can still occur, potentially impacting battery safety, longevity, and performance. A recent experimental study on the thermal safety of ASSLB packs showed that thermal runaway could occur and propagate in ASSLBs [5]. The thermal management is, therefore, a critical aspect of ASSLB design to mitigate these risks and optimize operational reliability according to investigate and understand

the thermal behaviours of ASSLB cells and their components.

The fundamental of heat generation in ASSLBs is like that in conventional LIBs. It is associated with ion/electron transport and electrochemical reactions between electrode and electrolyte during charging/discharging cycles. However, the absence of liquid electrolytes in ASSLBs alters the heat transfer mechanisms, making solid-phase conductivity critical for managing thermal gradients and maintaining interface stability. There is direct correlation between improved thermal conductivity and the mitigation of localized degradation mechanisms, such as phase changes and loss of solid-solid contact, which are unique to ASSLBs. Understanding thermal conductivity within ASSLBs is particularly important as it directly influences their ability to dissipate heat and prevent localized temperature variations. The composite cathodes of ASSLBs, comprising active materials (e.g., LiCoO_2) and solid electrolytes (e.g., $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$), present a heterogeneous microstructure where porosity, tortuosity, and component volume fractions govern thermal conduction. Localized thermal gradients in such microstructures can generate localized hotspots, exacerbate material degradation, reduce efficiency, and shorten cycle life. In the meantime, the structural changes of electrodes are the source of irreversible heat generation [6]. The uncontrollable heat generation might lead to thermal failure and also reduce the thermal stability [7]. Increasing the thermal

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conductivity can efficiently facilitate heat dissipation and prevent the possibility of localized-temperature hotspots [8]. Thus, studying the thermal conductivity of composite cathodes is essential to optimize their design for enhanced heat dissipation and stability under operational stresses.

Previous research efforts have been focused on optimizing the thermal conductivity of SE. For example, Yang et al. [9] reorganised the arrangement of poly ethylene oxide (PEO) and discovered that the thermal conductivity of crystalline PEO may be enhanced from $0.37 \text{ W m}^{-1} \text{ K}^{-1}$ to $60 \text{ W m}^{-1} \text{ K}^{-1}$. Bock et al. [10] doubled the lithium lanthanum zirconium oxide (LLZO) thermal conductivity from $0.24 \text{ W m}^{-1} \text{ K}^{-1}$ to $0.47 \text{ W m}^{-1} \text{ K}^{-1}$ with the compacting and sintering treatment. Meng et al. [11] demonstrated that high sintering temperatures can produce different thermal conductivity by the morphology change of solid electrolyte. In addition to the manufacturing process, the thermal conductivity of the electrode/electrolyte interface is also crucial. He et al. [12] discovered that the incorporation of poly acrylic acid (PAA) self-assembled monolayers (SAMs) at the LCO/PEO interface can significantly improve interface thermal conductivity.

Comparing to electrolyte, the thermal behaviour of cathode is relatively less studied. Typically, ASSLBs use composite cathodes by mixing SE and cathode active materials (CAM) particles to increase energy density and power density of ASSLBs. The characteristics of the composite cathode microstructure, including porosity, tortuosity, and the spatial distribution of active materials, directly influence their thermal conductivity. Due to the heterogenous nature of the cathode microstructure and the random distribution of particles, localized-temperature hotspots could be present as potential thermal hazards. Whilst conducting experiments on the thermal behaviour of the composite cathode is challenging, numerical simulations present an alternative way to understand the thermal behaviour of the composite cathode and microstructural effects [13]. Direct numerical simulations on the electrode microstructures present a promising route for investigating the effect of microstructures on electrode performances. Different modelling techniques, including finite volume method (FVM), finite element method (FEM), discrete element method (DEM), and lattice Boltzmann method (LBM), have been used to study the thermal behaviours of LIB electrodes.

FVM was extensively used in the thermal modelling of 3D electrode microstructures due to its ability to handle complex boundary conditions. The method divides the computational domain into a finite number of control volumes, applying conservation laws to each volume. A study used the OpenFOAM solver in 3D FVM showed that increasing the carbon-binder domain fraction in cathodes of LIBs notably improves the thermal pathways and enhance the effective thermal conductivity (ETC) [14]. Heat generation in LIB electrodes was first studied by using 3D FVM simulations [12,15] on detailed 3D electrode microstructures. Several heat sources were considered including electronic resistive heat, ionic resistive heat, contact resistive heat, reaction heat, entropic heat and heat of mixing. The results showed that the heterogeneous electrode microstructure led to complex and inhomogeneous spatial distribution of the heat generation rate in the LIB electrode during the discharge process, giving rise to higher heat generation than the theoretical model for the LCO cathode [16]. Instead of performing heat generation simulations directly on detailed 3D electrode microstructures, Mukherjee's group took an FVM-based approach, named microstructure-aware electrochemical-thermal coupled model (MAECT), to analyse how electrode microstructure of LIBs impacted heat generation, particularly in terms of ohmic, reaction, and reversible heating. Their simulation results on NMC cathodes revealed that increasing electrode porosity can reduce battery temperature due to a lower heat generation rate, and if the electrode porosity is low enough, the self-shutdown mechanism can also limit the battery temperature, especially for high current discharge. The group later extended the MAECT approach to study graphite anodes considering the effects of particle shape and packing fraction on transport limitations, heat generation, and battery fast discharge. [17]

FEM is another method widely used for thermal analysis in lithium batteries. It divides the domain into smaller elements and applies interpolation functions to solve the thermal equations. By using COMSOL to develop Multiphysics FEMs in the reconstructed LFP (LiFePO_4) microstructure of a half-cell system (i.e., with electrolyte and cathode), a battery life management study showed that temperatures above 45°C rapidly reduce the life of LIBs, while higher temperatures also pose safety concerns as the materials in the battery begin to degrade chemically and mechanically [18]. Thus, it is necessary to keep the temperature of LIBs not too high in order to prolong the battery life. Moreover, a simulation of COMSOL-based model for a LFP prismatic battery cell at different discharge rates showed that heat was produced more in the cathode than the anode, while lithium concentration was more uniform in the anode than in the cathode. The simulation results were validated by an experiment for charging a 20 Ah LFP prismatic Li-ion pouch battery cell. [19] Thus, the thermal property of cathode is a critical factor to explore for lithium battery performance optimization and thermal management.

The DEM is one of the off-lattice methods with no grid or mesh in the simulation domain [13], in which the domains are represented by Lagrange particles that move throughout the system due to the forces acting on them. This method models the interactions between individual particles within the electrode microstructures, which is particularly useful for understanding the thermal behaviour at the particle level. An open-source DEM code LIGGGHTS was used to generate electrode microstructures with varying microstructural properties (namely porosity, particle size, and electrode thickness) and simulate the overall thermal conduction of the electrodes. The study found that smaller particles increase thermal resistance and decrease heating rates, and higher porosity reduced the number of particle contacts, further lowering heat transfer efficiency. Optimal thermal conduction was achieved with thin electrodes made of large particles and low porosity [20].

LBM is a numerical approach that models fluid dynamics and heat transfer by simulating particle distributions on a lattice grid. Its application to thermal conductivity modelling in LIBs cathodes has shown promise, especially in capturing micro-scale heat transfer phenomena. LBM operates on a mesoscopic level, excels in managing complex geometries and heterogeneous microstructures found in cathodes without the need for complex mesh generation, which is often required in FVM and FEM. He et al. [21] used LBM to simulate the thermal diffusion process in the computer-generated 3D microstructure of graphite anode and predict the effective thermal conductivity. The results showed that the particle size has a certain effect on the effective thermal conductivity, but the effect is generally insignificant. Mehri et al. [22] investigates conjugate heat transfer in a microchannel with hydrophobic surfaces and nanofluids under a magnetic field by using LBM, showing nanofluids enhances heat transfer by up to 50 %, providing insights into designing efficient cooling systems for microscale applications. Dallahnejad et al. [23] explored that magnetic field notably influences pressure distribution and generates large vortices, while its effects on electric fields and ion concentrations are minimal.

Comparing these simulation methods, FVM is widely used for its accuracy in solving conservation laws (e.g., mass, energy) and its robustness in handling large-scale simulations of thermal and electrochemical processes in lithium batteries. However, it struggles with complex geometries like porous electrodes and requires computationally expensive meshing. FEM is highly accurate for structural and thermal analyses and is well-suited for multi-physics simulations, but its computational cost and difficulty in handling heterogeneous or transient systems limit its scalability. DEM excels in capturing particle-level mechanics, such as cracking or displacement during battery cycling, but is limited to mechanical behaviours and must be coupled with other methods for thermal or electrochemical simulations, increasing complexity. Unlike FEM and FVM, LBM inherently operates on a mesoscopic scale, offers distinct advantages, e.g., efficient handling of complex, heterogeneous microstructures without complex meshing,

high parallelizability for large-scale computations, and superior performance in simulating coupled heat and mass transfer at the micro-structural scale. But currently, LBM is limited for structural mechanics, lack of commercial software.

It should be noted that, although there are a number of numerical studies on the thermal behaviours of cathodes in conventional liquid electrolyte-based LIBs, thermal modelling for ASSLBs has been rare. Previous studies have identified the importance of thermal conductivity, which is beneficial to the thermal stability and battery performance during the whole working process. The heat transfer process in composite cathodes of ASSLBs is expected to be more complex than that of conventional LIBs, considering that electrolyte is now present as solid particles and the heat transfer process would be more sensitive to the microstructure details such as porosity, tortuosity, and the contacts between SE and CAM particles. A thorough understanding of the heat transfer process is imperative for developing reliable and robust ASSLBs.

In real-world applications, improved thermal management in ASSLBs translates to safer batteries with extended lifetimes and better performance. Efficient thermal conduction can mitigate thermal runaway risks, ensure uniform temperature distributions, and maintain the electrochemical integrity of active materials and interfaces. This connection between thermal conductivity and practical battery performance justifies the need for detailed investigations into the micro-structural factors influencing heat conduction.

Therefore, this paper reports a numerical modelling study of the heat transfer in composite cathodes for ASSLBs using an in-house 3D lattice Boltzmann (LB) model developed at the University of Surrey. The 3D LBM simulations of heat transfer will be directly performed on microstructures of composite cathodes generated using DEM modelling. The study aims to quantify the influence of parametric cathode microstructure design on the effective thermal conductivity, identify the effect of microstructure void region on the effective thermal conductivity, and provide guidance for the composite cathode microstructure design for ASSLBs. The methodology is outlined in Section 2. Section 3 presents the experiments and model validations. Comprehensive discussions about the different cathode microstructures are shown in Section 4. Section 5 summarises the key findings and demonstrates the optimised design. By systematically investigating key parameters such as porosity, volume ratio, tortuosity, and particle size, this work seeks to provide actionable insights into optimizing cathode microstructures for superior thermal management in ASSLBs. The findings from this research will offer valuable guidance for the design of next-generation batteries with improved safety and performance characteristics.

2. Methodology

2.1. Physical properties of the composite cathode in ASSLBs

The composite cathode in ASSLBs normally consists of solid electrolyte (SE) particles mixed with cathode active material (CAM) particles, forming an intricate microstructure, as shown in Fig. 1. There could be many different combinations of SE and CAM composites, depending on the choices of SE and CAM materials. SE materials could include

polymer-based (e.g., PEO), oxides-based (e.g., LLZO, LLZTO, LATP), and sulphides-based (e.g., LGPS, LiSiPS, LiPS); whilst CAM materials could include LCO, LFP, NMC, and LMO.

The physical properties of the composite cathode formed using LCO as CAM and LLZO as SE are listed in Table 1. The thermal diffusivity of each material can be obtained using Eq. (1):

$$\alpha_T = k / (\rho C_p) \quad (1)$$

where α_T is the thermal diffusivity, k indicates the thermal conductivity, ρ represents the density, and C_p is the specific heat capacity.

2.2. Digital structure generation

Creating the digitalised structure is the essential step in modelling and simulations for the performance analysis. The discrete element method (DEM) is a well-established numerical approach for studying particle behaviours including particles interactions and dynamic flows [20,28]. Besides modelling the particle dynamic flows, DEM is also widely used to simulate the structures of agglomerations consisting of particles, such as electrode cells in electrochemical energy devices [29,30]. One of the main advantages of DEM is its ability to simulate random particle coordination, which plays a crucial role in ensuring the DEM-created porous structure has a similar distribution uniformity of solid components to the real-world composite cathode. DEM employs stochastic approaches to distribute particles in a manner that mimics real-world granular packings, ensuring statistical uniformity [31]. Applying periodic boundary conditions in DEM ensures that the generated structure does not have artificial edge effects that could disrupt homogeneity. And the 3D coordination number (i.e., the number of contacts a particle has with its neighbours) is randomized to reflect natural granular assemblies, avoiding localized clustering. This capability is vital for various applications, including composite cathode manufacturing and porous media analysis. Therefore, an in-house DEM code is used in this work to generate different structures of the composite cathodes.

Since the computational cost of modelling the whole cathode is impractical, it is essential to select an appropriate representative elementary volume (REV), which is defined as the smallest volume over which a measurement (e.g., temperature, thermal conductivity) can be averaged to provide a representative value for the entire material. REV provides a balance between computational efficiency and accuracy, reducing the need for excessive computational resources while still capturing the essential thermal dynamics. According to the past investigation, the characterized volume of the electrode for REV should correspond to a cube with a minimum length of 7.5 times the diameter of the particle [32]. Therefore, this article selects the cathode samples of $80 \mu\text{m} \times 80 \mu\text{m} \times 100 \mu\text{m}$ as the REV since $80 \mu\text{m}$ is large enough for the largest particle diameter of $10 \mu\text{m}$.

Both LCO and LLZO grains are set as spherical-shaped in the DEM model. While the LCO grain is normally a layered oxide material with a hexagonal close-packed structure, we opted for spherical-shaped grains in our simulation for the following reasons: Firstly, spherical grains are often used as a standard approximation for microstructural modelling in porous media. While grain shape can influence local heat conduction pathways, this assumption of spherical grains provides a baseline understanding on how the structure parameters affect (e.g., porosity, volume fraction, tortuosity, and particle size) the effective thermal conductivity (ETC) and enables comparison of ETCs for different composite cathodes by controlling some structure parameters. Secondly, in practical cathode designs, LCO particles often undergo processing steps such as ball milling, which result in deviations from the original hexagonal shapes. The spherical assumption enables efficient packing algorithms in the DEM and facilitates faster convergence of simulations. This trade-off between geometric fidelity and computational tractability helped to ensure the feasibility of simulating a wide range of

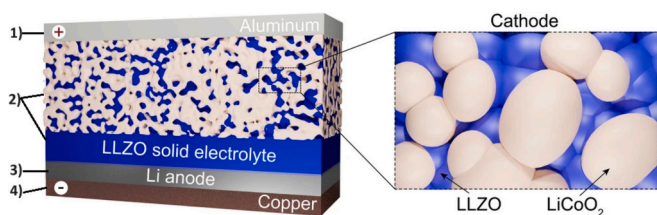


Fig. 1. Schematics of the SSBs: 1) Aluminium foil as cathode current collector; 2) a cathode consists of active material (LCO) and solid electrolyte (LLZO); 3) a lithium metal anode; 4) an electrolytic copper foil. [24].

Table 1
Physical properties of cathode materials

	Thermal conductivity k (W m ⁻¹ K ⁻¹)	Density ρ (g cm ⁻³)	Specific heat capacity C_p (J kg ⁻¹ K ⁻¹)	Thermal diffusivity α_T (m ² s ⁻¹)
LCO (LiCoO ₂) [25]	5.40	5.06	715.42	1.49×10^{-6}
LLZO (Li ₇ La ₃ Zr ₂ O ₁₂) [10,26,27]	0.46 (0.47 ± 0.01)	5.11	927.19	1.00×10^{-7}

microstructural configurations. Furthermore, experimental studies showed that it was possible to synthesize spherical LCO materials [33] and that these spherical LCO materials can enhance Li-ion battery performance compared to the hexagonal ones [34]. Therefore, the use of spherical grains in the model can be seen as an approximation of these processed geometries. For more realistic representation of the composite cathode microstructure, future work could focus on integrating hexagonal or other anisotropic particle shapes into the DEM framework to study their impact on heat conduction pathways and ETC. This would involve developing advanced algorithms to handle the orientation, packing, and contact mechanics of non-spherical grains.

The structure generation process of the ASSLBs composite cathode by the DEM is shown in Fig. 2. It consists of particles generation, deposition and compression processes. Certain numbers of red particle 1 (represent LCO) and blue particle 2 (represent LLZO) were randomly generated within a 3D container (80 μ m × 80 μ m × 430 μ m). Particles deposited in the bottom of the container under the gravity and finally reached the steady state. Afterwards, a top plate was applied and moved downwards with a constant velocity of 0.003 m/s. To obtain the required overall porosity, the particle bed was compressed into 80 μ m × 80 μ m × 100 μ m. In this paper, different microstructures were generated based on different particle sizes, volume ratios, overall porosities, and tortuosities. The volume ratios and overall porosities of cathodes are calculated via the grid statistics of the cathode 3D microstructure mesh on the MATLAB platform, and the tortuosity is derived by TauFactor [35].

The DEM-generated cathode structures are digitalised into 3D matrices with different numbers denoting related composites (i.e., “0” for pores, “1” for LCO, “2” for LLZO) and then input to the 3D LB heat transfer model to investigate the effect of microstructure on the heat conduction at equilibrium states. The mesh resolution (grid size) for the generated cathode structures is 1 μ m, and the size of cathode sample is 80 μ m × 80 μ m × 100 μ m as shown in Fig. 2, thus the sample mesh dimension is 80 × 80 × 100 (the total grids number of a sample is

640,000).

In this study, our in-house DEM computer programme [36–38] was adapted for thermomechanical analyses. Each particle is considered as a distinct element, influenced by both contact and gravitational forces. The translational and rotational movements of a primary particle are determined by the following equations:

$$m_i \frac{d\mathbf{v}_i}{dt} = \left(\sum \mathbf{F}_{n,ij} + \sum \mathbf{F}_{s,ij} \right) + m_i \mathbf{g} \quad (2)$$

$$I_i \frac{d\boldsymbol{\omega}_i}{dt} = \sum \mathbf{M}_{ij} = \sum (\mathbf{F}_{s,ij} \times \mathbf{r}_{ij}) \quad (3)$$

where m_i , I_i and $\mathbf{M}_{ij}$ are the mass, moment of inertia, and torque by tangential force of particle i ; $\mathbf{v}_i$ and $\boldsymbol{\omega}_i$ denote the translational velocity and angular velocity of particle i ; $\mathbf{F}_{n,ij}$ and $\mathbf{F}_{s,ij}$ are the normal force and tangential force between two contacting particles i and j ; $\mathbf{g}$ is the gravitational acceleration; $\mathbf{r}_{ij} = r_i \mathbf{n}_{ij}$, where $\mathbf{n}_{ij}$ is the unit vector connecting the centre of particles i and j ; and t represents the time.

This study used the classical Hertz-Mindlin-Deresiewicz models to determine the normal and tangential contact interactions for frictional-elastic particles without adhesion and electrostatic forces. Based upon the Hertz theory, the magnitude of the normal contact force F_n between two particles is calculated as [38,39]:

$$F_n = \frac{4}{3} E^* R^{*1/2} \alpha^{3/2} \quad (4)$$

where α is the overlap between particles, E^* is the equivalent elastic modulus, and R^* is the equivalent radius, which are defined as:

$$\frac{1}{E^*} = \frac{1 - \nu_i^2}{E_i} + \frac{1 - \nu_j^2}{E_j} \quad (5)$$

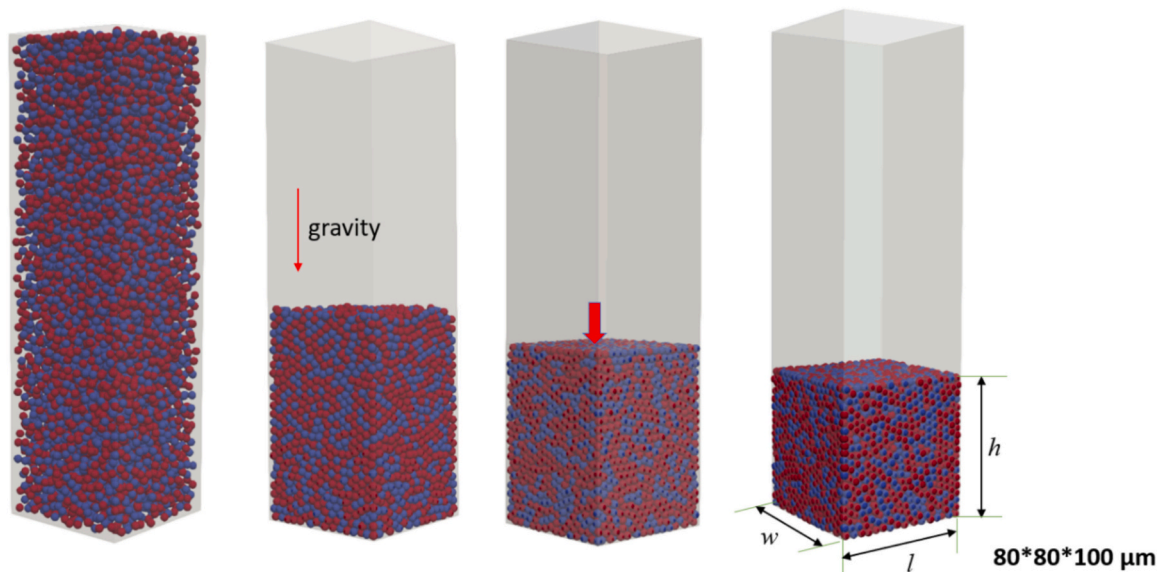


Fig. 2. Cathode structure generation by DEM, red and blue particles represent LCO and LLZO. Certain numbers of red particle 1 (represent LCO) and blue particle 2 (represent LLZO) were randomly generated within a 3D container (80 μ m × 80 μ m × 430 μ m), in which the porosity is 8.1 %, volume ratio of LCO and LLZO is 1:1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$$\frac{1}{R^*} = \frac{1}{R_i} + \frac{1}{R_j} \quad (6)$$

where E_i , E_j , ν_i , ν_j , R_i and R_j are, respectively, the elastic moduli, Poisson's ratios, and radii of the contacting elements i and j . When the neighbouring element is a wall, its radius is considered as infinity.

The contact force in the tangential direction is determined by the Mindlin and Deresiewicz model [40], which is updated incrementally. The incremental tangential force ΔF_s , which results from the incremental tangential displacement Δa_t , depends on both the incremental normal force ΔF_n and the historical loading pattern, i.e.

$$\Delta F_s = 8G^* r_c \theta_k \Delta a_t + (-1)^k \mu \Delta F_n (1 - \theta_k) \quad (7)$$

where μ donates the coefficient of friction, r_c is the radius of the contact area between elements given as [41]:

$$r_c = \sqrt{\alpha R^*} \quad (8)$$

And G^* is the equivalent shear modulus expressed as:

$$\frac{1}{G^*} = \frac{1 - \nu_i^2}{G_i} + \frac{1 - \nu_j^2}{G_j} \quad (9)$$

where G_i and G_j are the shear moduli of elements i and j .

For no-slip conditions:

$$\theta_k = 1 \text{ if } |\Delta F_s| < \mu \Delta F_n \quad (10)$$

For partial-slip conditions:

$$\theta_k = \sqrt[3]{1 - \frac{F_s + \mu \Delta F_n}{\mu F_n}} \text{ if } k = 0 \quad (11)$$

$$\theta_k = \sqrt[3]{1 - \frac{(-1)^k (F_s - F_s^c) + 2\mu \Delta F_n}{2\mu F_n}} \text{ if } k = 1, 2 \quad (12)$$

where $k = 0, 1, 2$ denote loading, unloading, and reloading paths, respectively, and F_s^c represents the historic tangential force at the load reversal points during loading or reloading. Therefore, the total tangential contact force is calculated as:

$$F_s = \begin{cases} \sum \Delta F_s & \text{if } F_s \leq \mu F_n \\ \mu F_n & \text{if } F_s > \mu F_n \end{cases} \quad (13)$$

While the DEM approach is effective for modelling pristine cathode structures, it has inherent limitations that restrict its ability to directly simulate microstructural degradation during cycling. Key degradation phenomena, such as CAM cracking, loss of contact between SE and CAM, or the formation of micro voids, occur due to mechanical and chemical stresses during charge and discharge. These dynamic processes cannot be directly modelled with DEM, as the method does not account for stress propagation, fracture mechanics, or interfacial dynamics. These changes in microstructure can significantly alter the pathways for heat conduction, increasing thermal resistance and creating localized hot-spots. Such degradation not only affects thermal conductivity but also exacerbates performance degradation and safety risks over time. Although these effects cannot be directly simulated using the DEM structure generation method employed in this study, future extensions of this work could incorporate cycling-induced changes into the simulation framework to better understand scenarios of heat conduction in aged ASSLB cathodes. The scope of this study, focused on pristine cathode microstructures using a controlled and simplified framework, offer a baseline understanding of thermal behaviour in ASSLBs

2.3. The heat transfer model

2.3.1. The heat conduction lattice Boltzmann model

This paper used three-dimensional (D3Q19) lattice Boltzmann (LB) model to investigate heat conduction in different cathode microstructures. The classical LBM is based on the advection-diffusion equation, including unsteady thermal diffusion, square-cavity convective heat transfer and natural convection [42]. To simplify the model, the viscous heating, dufour effect, and soret effect are ignored in the current study. Then the LBM [43] can be expressed as:

$$\frac{\partial T}{\partial t} + \nabla \cdot (uT) = \nabla \cdot (\alpha_T \nabla T) + S \quad (14)$$

where T represents temperature, $\partial T / \partial t$ is the temperature partial derivative with respect to time t , $\nabla \cdot (uT)$ is the convective term, S is the heat source term, α_T is thermal diffusivity and can be calculated by Eq. (1). The source term is defined by the heat generation rate $\dot{q}$:

$$S = \frac{\dot{q}}{\rho c_p} \quad (15)$$

According to Guo et al. [44], the evolution of the temperature field ought to be described by another temperature distribution function:

$$f_a^{eq} = w_a T \left[1 + \frac{c_a \cdot u}{c_s^2} \right] \quad (16)$$

where α denotes the number of lattice vectors, c_a indicates the lattice speed, w_a is weight coefficient and c_s is sound speed. Since the composites of ASSLBs are all solid and no flow and convection within cathodes, the velocity (u) is zero. Therefore, Eq. (4) is simplified into:

$$f_a^{eq} = w_a T \quad (17)$$

τ_T is the dimensionless relaxation time. The discrete evolution equation of the temperature field is [45]

$$f_a(\mathbf{r} + \mathbf{e}_a \Delta t, t + \Delta t) - f_a(\mathbf{r}, t) = -\frac{1}{\tau_T} [f_a(\mathbf{r}, t) - f_a^{eq}(\mathbf{r}, t)] + \Delta t w_a S \quad (18)$$

The temperature distribution T is defined by the TDF as

$$T = \sum_a f_a \quad (19)$$

According to the Chapman-Enskog procedure by Guo et al. [44], the relation between the thermal diffusivity ($\alpha_T = \nu_t / Pr$) and relaxation time τ_T is

$$\alpha_T = c_s^2 [\tau_T - 1/2] \Delta t \quad (20)$$

Fig. 3 illustrates the major process for LBM heat conduction model, whose process during iterations can be summarized to several key steps (1)–(6) as follows. The boundary conditions and relaxation time settings will be discussed in detail in next subsections.

1. Initialize T , f_a and f_a^{eq}
2. Collision step (relaxation): calculate the updated distribution function $f_a(\mathbf{r} + \mathbf{e}_a \Delta t, t + \Delta t)$ using Eq. (18) at each iteration step and for each lattice node. Since the thermal diffusivities of different constituent materials in cathode are not the same, the simulation area should be divided into separated regions by the phases of materials and applied their respective relaxation time by the multi-time scale implementation (see more details in Section 2.3.3) for evolution equations.
3. Streaming step: move $f_a \rightarrow f_a^*$ in the lattice direction of $\mathbf{e}_a$, in which f_a^* is the updated particle distribution function at the adjacent nodes to f_a . Since the simulation area has several separated phases, the streaming step of different cathode material phases are launched in their simulation regions separately during every iteration step.

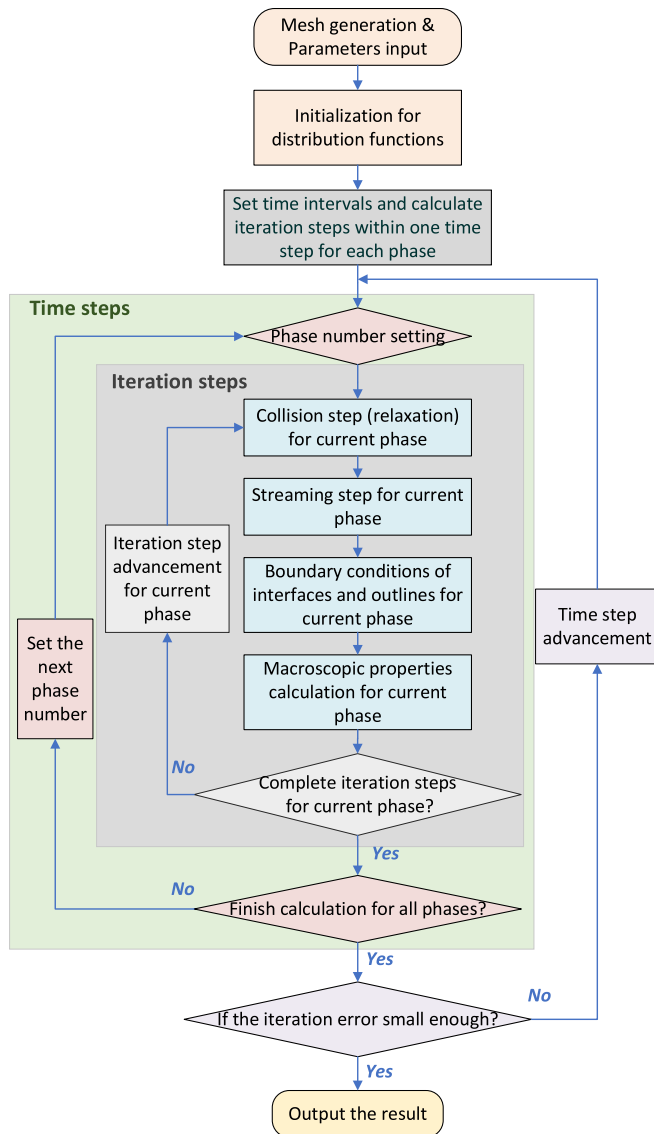


Fig. 3. Flow chart of LBM for heat conduction model: step 1) Initialize T , f_α and f_α^{eq} ; step 2) Collision step (Eq. (18)): calculate the updated distribution function f_α ; step 3) Streaming step: move $f_\alpha \rightarrow f_\alpha^*$; step 4) Apply appropriate boundary conditions; step 5) Compute macroscopic properties T from f_α^* (Eq. (19)).

4. Boundary conditions: Apply appropriate boundary conditions to simulate walls, inlets, outlets, or interfaces within the simulated cathode. Because the collision and streaming steps are launched in separated phases respectively, the heat conduction processes between the connected material phases are calculated by the interface boundary conditions (will be discussed in Section 2.3.2).
5. Compute macroscopic properties T from f_α^* using Eq. (19) at each. The temperature field is used to analyse the heat transfer and for visualization.
6. Repeat steps (2)–(5) with time step advancement until a convergence criterion based on changes in simulated temperature field.

The assumption of zero velocity in the LB model is valid for steady-state heat conduction but may lead to inaccuracies when simulating transient thermal scenarios, particularly in systems like LIBs where heat generation and dissipation are complex and influenced by convective effects.

Convection can play a significant role in transient thermal scenarios, especially during high-current operations or thermal runaway

conditions. For instance, Zhou et al. [46] identified that convection significantly influences heat dissipation efficiency of LIBs in densely packed configurations, affecting overall battery performance and life-span. This study underlines the importance of including convective terms when addressing heat dissipation challenges in transient scenarios. Additionally, Jiang et al. [47] used LBM to highlight that improved convective heat transfer on LIB surfaces mitigates thermal runaway effects, emphasizing the interplay of convection and heat generation during transient conditions. Furthermore, Li et al. [48] modelled thermal runaway in high-energy battery packs by integrating natural convection effects, and underscored the necessity of accounting for convection to capture the thermal dynamics accurately during extreme conditions in LIBs.

However, the unique configuration of ASSLBs, characterized by the absence of liquid electrolytes and the vacuum state within electrode pores due to the manufacturing process, fundamentally alters the thermal transport mechanisms compared to traditional LIBs, and thus, there is no medium in electrode pores (which are in near vacuum state) for fluid flow or gas-phase convection to transport heat, leaving heat conduction through solid materials as the sole internal heat transfer mechanism. The LB model for heat transfer of ASSLBs must exclude fluid dynamics elements associated with flow-induced convection. Instead, the focus shifts to accurately modelling conduction and the anisotropic thermal properties of solid components.

2.3.2. Boundary conditions

Fig. 4 and Table 2 show the schematic of the computational domain and boundary conditions of the current study for solid-state composite cathode samples. At the top of the surface (marked in blue on Fig. 4), a constant heat flux of $200,000 \text{ W/m}^2$ is given and a temperature of 343 K is applied to the bottom surface (marked in red). All the other walls are set as adiabatic (also as vertical surfaces, mark in grey). At the LCO/LLZO (and air/steel in Section 3.2) contacted interface, since the specific heat capacities c_p of those two materials are different, which are marked as material 1 and material 2, the setting of interface boundary in Table 2 is discussed below.

The current study employs simplified outline boundary conditions (Dirichlet at the bottom surface and Neumann at the top) to systematically investigate the effects of microstructural parameters on effective thermal conductivity. This approach aligns with common practices in literatures [21] and provides a controlled framework for parametric analysis. Additionally, radiative and convective heat transfer mechanisms were excluded for the interface and outline boundary conditions of the LB model for composite cathode samples. Because heat transfer in composite cathodes is dominated by conduction through the solid phases (active materials LCO and solid electrolytes LLZO) and interfacial regions. Radiative and convective mechanisms typically play negligible roles, especially under the low to moderate temperature conditions relevant to solid-state battery operation. This is because the pores within the composite cathodes of ASSLBs would contain minimum gases (i.e. near vacuum condition) after being fabricated and assembled in glovebox.

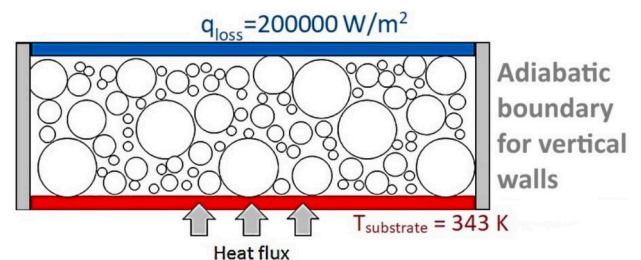


Fig. 4. Boundary conditions of temperature fields in cathode microstructure samples.

Table 2
Outline and interface boundary condition (BC) settings for cathode samples

Boundary classification	Surface of sample	Boundary type	Value for setting
Outline boundary	Top surface	Neumann BC	Heat flux $q_{\text{loss}} = 200,000 \text{ W/m}^2$
	Bottom surface	Dirichlet BC	Bottom temperature $T_{\text{substrate}} = 343 \text{ K}$
	Vertical surfaces	Adiabatic BC	
Interface boundary	Material 1 side surfaces at material 1/2 interface	Dirichlet BC	T_e at related interface grids of material 1/2
	Material 2 side surfaces at material 1/2 interface	Dirichlet BC	T_e at related interface grids of material 1/2
	Material 1 side surfaces at material 1/pores interface	Adiabatic BC	N/A
	Material 2 side surfaces at material 2/pores interface	Adiabatic BC	N/A

While these conditions do not capture the full complexity of real-world thermal behaviours near interfaces, they are appropriate for the pristine microstructures considered in this work. Future studies should extend this work by incorporating more realistic boundary conditions to account for complex thermal behaviours near interfaces, e.g., thermal boundary resistance, radiative heat transfer or convective heat losses at the cathode's surface and non-uniform heat fluxes. These developments, coupled with multi-scale modelling frameworks, would enable a more comprehensive understanding of heat conduction in real-world composite cathodes.

It should be noted that, at the LCO/LLZO interface, since the specific heat capacities c_p of those two materials are different, the setting of interface boundary is important. Detecting the interface boundary of two phases in the LBM involves identifying the regions where the two phases are in contact. Interface boundary detection procedure of the porous cathode microstructure is summarized in the following steps and shown as Fig. 5.

It is also acknowledged that the current interface boundary condition setting assumes thermal equilibrium at the LCO/LLZO (active material / solid electrolyte) interface simplifies the simulation and does not fully capture the complexities associated with real-world thermal interactions at material interfaces. However, the exact values of interface thermal resistance for LCO and LLZO have not been consistently reported in peer-reviewed literatures. This lack of universally accepted experimental data makes it challenging to directly incorporate precise interface resistance values into our simulations. Measuring the thermal resistance at the nanoscale LCO/LLZO interface in ASSLBs is further complicated by fabrication variability, such as surface roughness and bonding quality, and the lack of universally applicable experimental methods or values. Additionally, interface resistance is highly sensitive to material-specific properties (e.g., doping of LLZO or LCO surface treatments) and manufacturing processes (e.g., sintering temperature, pressure, and interface additives), making generalization across configurations impractical. Explicitly incorporating this resistance into simulations would require scarce, inconsistent data and significantly increase computational demands. While this approach does not explicitly capture nanoscale interfacial resistance, it provides robust insights into the dominant factors affecting effective thermal conductivity (ETC), such as porosity, tortuosity, and component volume fractions. Designing an experiment to measure the specific nanoscale LCO/LLZO interface thermal resistance can be future research, in which the experimental resistance values can be introduced into our LB model for more accurate simulation results.

After the interface boundary condition detecting process, it is necessary to set the accurate boundary values. Since grid size should be

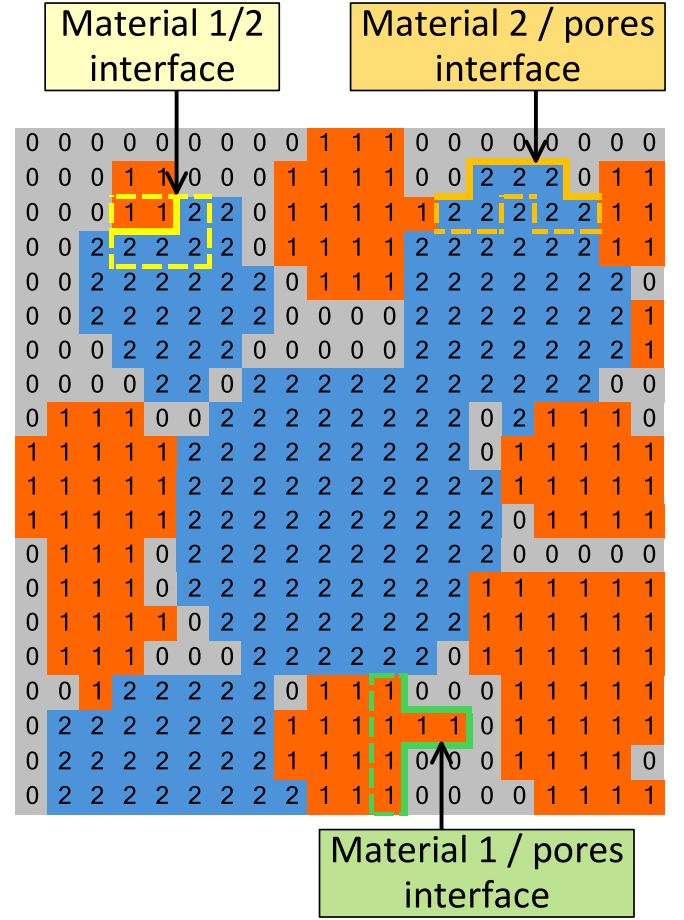


Fig. 5. Interface boundary detection schematic and steps: 1) Digitalize the cathode microstructure, LCO for 1, LLZO for 2 and pore for 0; 2) locate the interface boundary grids; 3) identify three interface types, LCO/LLZO, LLZO/pore, and LCO/pore; 4) apply the correct boundary condition settings.

set very small compared to the structure size, the temperature at the material 1/2 grids interface during one time step is assumed to reach local thermal equilibrium temperature T_e within one iteration [49,50]. The hypothesis of local thermal equilibrium is often used for heat transfer in porous media when the heating or cooling procedure is not rapid [50]. The boundary condition settings of interface grids should be set as the Dirichlet boundary with thermal equilibrium temperature T_e of two contacted material 1/2 grids instead of the arithmetic average temperature. The interface thermal equilibrium temperature T_e for material 1/2 contacted grids is described in Fig. 6, which is introduced into the LB model for the interface thermal boundary condition as the Dirichlet boundary condition. We previously discussed the related LB boundary equations in detail [51]. The following equations are used for the interface settings.

$$\Delta Q_i = c_{p,i} m_i (T_i - T_e), \quad (21)$$

$$\begin{aligned} m_i &= \rho_i dx^3 \\ \Delta Q_1 + \Delta Q_2 &= 0 \end{aligned} \quad (22)$$

$$T_e = \frac{c_{p,1}\rho_1 T_1 + c_{p,2}\rho_2 T_2}{c_{p,1}\rho_1 + c_{p,2}\rho_2} \quad (23)$$

The local thermal equilibrium (LTE) assumption used in this study simplifies simulations by treating interface temperatures as equilibrated within each time step. While this approach is validated against experimental data and is appropriate for pristine microstructures, it may overlook thermal boundary resistance arising from temperature

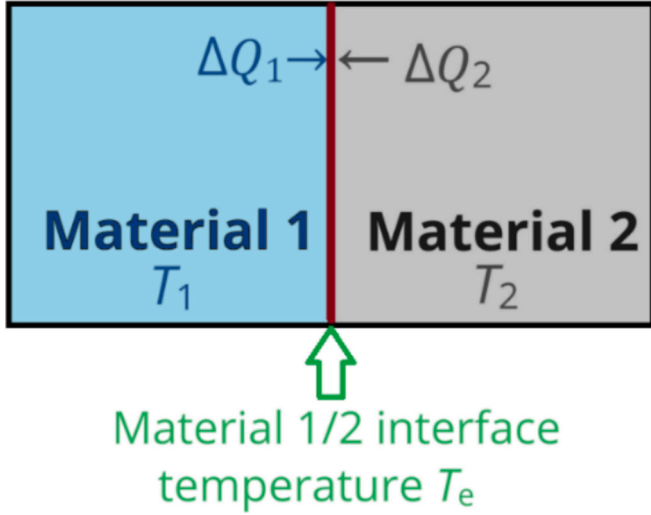


Fig. 6. The interface thermal equilibrium temperature T_e for material 1/2 contacted grids.

gradients between phases (e.g., active material and electrolyte). Such resistance can significantly affect heat transfer, particularly in highly heterogeneous systems or degraded structures with poor interfacial contact. To address this, future extensions of the model could incorporate interface-specific thermal resistance or multi-scale approaches to capture interfacial phenomena more accurately.

2.3.3. Relaxation time

The relation between the relaxation time τ_T and the relaxation coefficient ω is $\omega = 1/\tau_T$. It is necessary to meet the criteria $0 < \omega < 1$ in single relaxation time (SRT) for collision step. For the complex geometrical simulation region, the iterations could be unstable and divergent when $\omega < 0.4$ and $\omega > 0.95$ according to the tests for the LCO and LLZO cathode microstructures. The thermal diffusivities of LCO ($\alpha_{T,LCO}$) and LLZO ($\alpha_{T,LLZO}$) are significantly different (as shown in

Table 1), $\alpha_{T,LCO}$ is >10 times larger than $\alpha_{T,LLZO}$. The relaxation coefficients ω need to be set obviously different when the length and time scale are the same for LCO and LLZO regions. It is impossible to set $0.4 < \omega < 0.95$ for both LCO and LLZO. (e.g., if $\delta x = 1 \times 10^{-6}$ m, $\delta t = 5 \times 10^{-7}$ s, thus $\omega_{LCO} = 0.1058$, $\omega_{LLZO} = 0.9074$), and the optimum relaxation time can be calculated by inserting $\omega = 1/\tau_T$ into Eq. (20).

To increase the stability of LB models, Servan-Camas et al. [52] proposed the two-relaxation-time collision operator for (LTRT) for the ADE-based LB models, but this increase the complexity of evolution equations and is hard to apply for the simulation region with complex geometrical boundaries. Therefore, the multi-time scale implementation in SRT is put forward to solve this problem. The multi time scale approach is appropriate for heat and mass transfer, in which different components are divided to different sub-regions with keeping the same length scale but setting different time scales, other parameters related to time scale need to be reset in independent sub-regions. The collision step of each sub-regions is still in SRT.

In the digitalized cathode microstructure, the sub-regions for LCO and LLZO are set in different time scales to keep the relaxation coefficients ω of those sub-regions in appropriate values ranges as $0.4 < \omega < 0.95$ and avoid iteration divergence. And the lattice scale parameters in each region should be converted to physical scale (SI unit) parameters for calculating the LCO/LLZO interface boundary conditions. As a result, to ensure the convergence, the relaxation coefficients of LCO and LLZO are set as 0.6177 and 0.7984, respectively. The time step is 2.5×10^{-6} s, where one time step includes 10 iteration steps for LCO sub-region and one iteration step for LLZO sub-region.

2.3.4. Calculation of the effective thermal conductivity

The effective thermal conductivity (ETC) is an important indicator to describe the thermal conductivity of a cathode with complex structures. The ETC with a particular direction x_i can be calculated as [53]

$$k_{eff,x_i} = \frac{q_{loss}}{\left(\frac{\partial T}{\partial x_i}\right)} \quad (24)$$

Eq. (24) in numerical simulations can be written as $k_{eff,x_i} = q_{loss} \Delta x_i / \Delta T$, set another equation not in the content Δx_i is the length of grids along the direction of heat transfer, ΔT is the difference between the equilibrium temperature of two sides of grids in Δx_i .

For modelling the packed particle model in Section 3, the heat loss flux q'_{loss} to the environment from Eq. (25) can replace q_{loss} in Eq. (24) to calculate a convective losses [54] for Robin boundary. For the packed particle model, the top surfaces of cathodes can be deemed as a horizontal plate for natural convection, in which the correlation between the average Nusselt number Nu and Rayleigh number Ra can be selected for laminar [55] and turbulent flow [56] as Eq. (26). In Eq. (27), Gr is the Grashof number, $Pr = \nu_f / \alpha_T$ is the Prandtl number, g is gravity acceleration, T_s is the surface temperature, T_∞ is the ambient temperature, β is the thermal expansion coefficient that equals to $1/T_f$, the film temperature $T_f = 1/2(T_s + T_\infty)$, L_c is the characteristic length, ν_f is the fluid kinematic viscosity, α_T is the thermal diffusivity. The characteristic length for a horizontal upper surface of a hot plate can be obtained by Eq. (28), in which A_s is the surface area, p is the perimeter. According to the Eq. (29) of Nusselt number Nu , the local thermal convection coefficient h can be calculated.

$$q'_{loss} = h(T_s - T_\infty) \quad (25)$$

$$Nu = \begin{cases} 0.5Ra^{1/4} & (Ra < 160) \\ 0.45Ra^{1/4} & (160 \leq Ra < 1 \times 10^4) \\ 0.54Ra^{1/4} & (1 \times 10^4 \leq Ra < 1 \times 10^7) \\ 0.15Ra^{1/3} & (1 \times 10^7 \leq Ra < 1 \times 10^{11}) \end{cases} \quad (26)$$

$$Ra = GrPr = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu_f \alpha_T} \quad (27)$$

$$L_c = A_s/p \quad (28)$$

$$Nu = hL_c/k \quad (29)$$

2.3.5. Convergence evaluation

To ensure accuracy of the results, firstly, the ETC should be calculated towards the temperature field of an iteration converged heat conduction simulation. The iteration convergence in heat conduction simulations can be determined through the L^2 -norm-based method, which measures the overall change in the temperature field. L^2 -norm is calculated in Eq. (30), where T^n and T^{n+1} are the temperature fields at successive iterations ("n" and "n+1" denote n-th and (n+1)-th iteration, respectively), N represents the total amount grids of the temperature field.

$$\|\Delta T\|_2 = \sqrt{\frac{\sum_{x,y,z} (T^{n+1}(x,y,z) - T^n(x,y,z))^2}{N}} \quad (30)$$

The convergence criterion of L^2 -norm is $\|\Delta T\|_2 < \epsilon$, in which ϵ is a predefined small tolerance value, which is evaluated by monitoring the steady-state temperature field across the computational domain. We investigated the relationship between the simulated results (ETCs), and iteration errors as the computational steps increased. By systematically monitoring the L^2 -norm, we established that the convergence criterion ϵ could reliably be set at 1×10^{-7} K. The choice of this stringent convergence criterion was validated through careful observation of the temperature distribution, which demonstrated stability and consistency

once the L^2 -norm fell below the defined threshold.

Secondly, the numerical stability was verified by the grid independence validation through a systematic mesh refinement study, which confirmed that the ETC results were independent of the mesh size beyond a specified resolution. Locally refined meshes were used to resolve complex interfaces, particularly in high-porosity systems. This validation involved testing a cathode sample with dimensions of $80 \mu\text{m} \times 80 \mu\text{m} \times 100 \mu\text{m}$, subjecting it to five different grid resolutions: a coarse mesh ($80 \times 80 \times 100$, $1 \mu\text{m}$ grid scale), a loose mesh ($40 \times 40 \times 50$, $2 \mu\text{m}$ scale), an intermediate mesh ($60 \times 60 \times 75$, $1.333 \mu\text{m}$ scale), a refined mesh ($120 \times 120 \times 150$, $0.667 \mu\text{m}$ scale), and a fine mesh ($160 \times 160 \times 200$, $0.5 \mu\text{m}$ scale). The porosity was set at 21.1 %, the particle sizes for LCO and LLZ were $5 \mu\text{m}$ and $10 \mu\text{m}$ respectively, and the volume fraction of LCO to LLZ was 1:1. Simulations used these parameters to determine temperature distributions along the z-direction, the primary axis for heat conduction. Comparative analysis of the temperature profiles confirmed result convergence at grid scales $\leq 1 \mu\text{m}$ (grid sizes $\geq 80 \times 80 \times 100$). This finding validated that the numerical solution for the coarse mesh ($80 \times 80 \times 100$, $1 \mu\text{m}$ scale) was unaffected by further mesh refinement and deemed sufficient for the simulation. These rigorous approaches underpin the reliability of the simulation results for assessing thermal conduction in ASSLBs.

3. Model validation

3.1. Experiment design

Comparisons with experimental data from real ASSLB cathodes could provide additional validation and strengthen the discussion. However, direct experimental measurements on real cathode microstructures in ASSLBs remain challenging and often unreliable, and the derivation of the effective thermal conductivity (ETC) of real cathode microstructures in ASSLBs is difficult with the current experimental techniques. Techniques like thermal imaging have been used to obtain temperature distribution within batteries, but often at the pack/cell level, not to the particle level with the details of 3D microstructures [57]. Although X-ray tomography techniques can obtain 3D microstructures at the particle level, they are not equipped to reveal temperature distributions. Thermocouples and advanced sensors have been used to measure temperatures on the surfaces of the batteries and the electrodes, but cannot penetrate into the electrodes to provide temperature distribution along the electrode thickness, without interfering battery performance. We have performed a comprehensive review on experimental and computational techniques for thermal analysis for batteries [4]. So far, we have not been made aware of any experimental techniques that can measure the spatial temperature distributions and effective thermal conductivities within 3D electrode microstructures for batteries.

Given these constraints, we have adopted the packed particles approach for model validation. Although the packed particles do not directly match the 3D cathode microstructures of ASSLBs, it allows us to map the temperature distribution along the thickness of the packed particle bed, which we can use to compare directly with our simulation results on the packed particle bed. This packed particle approach allows us to count for material properties, structural dynamics, and interfacial effects in both experimental measurement and computational simulations, providing an indirect but strong validation that our model is able to capture the heat conduction behaviours at the particle level.

To validate the model of the heat conduction in a packed porous structure, a heat transfer experiment is performed on packed particles. The packed particles model is commonly used to understand and predict the behaviour of granular materials (GMs), which could represent collections of discrete solid, macroscopic particles, e.g., a packed bed [58], graphite, solid state cathode in ASSLBs. Previous studies have shown the feasibility of the packed bed experiments for model validations [58–60]. Following the same practice, an experiment is designed to validate the LB model, as shown in Fig. 7. A thick plastic tube (inner diameter: 41

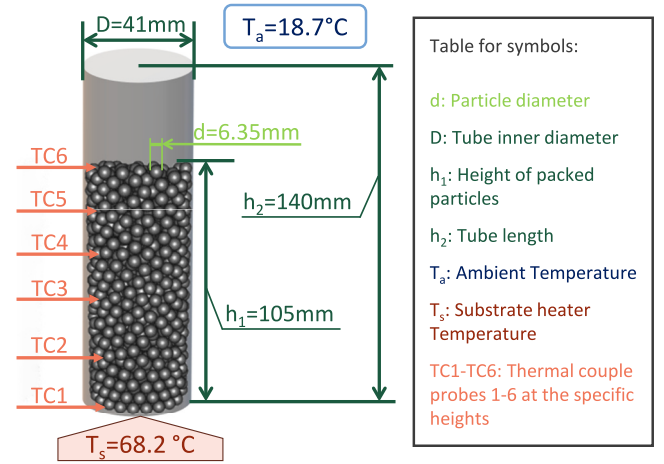


Fig. 7. Diagram of heat transfer experiment for packed particles.

mm, length: 140 mm) is selected to contain packed steel particles (of 6.35 mm diameter spheres, height of accumulated packed particle: 105 mm). The thermal insulation material is used to cover the tube for a better thermal insulation effect. The particles are heated at the bottom with a constant temperature $T_s = 341.35 \text{ K}$ (68.2°C). The top side is exposed to the environment with the ambient temperature $T_a = 291.9 \text{ K}$ (18.7°C) and emits the heat to the ambient by natural convection. The other sides are adiabatic due to the thermal insulation.

During the experiment, the thermocouple probes are used to record the temperatures in different depths every minute until the heat transfer process reaches the equilibrium state. Six thermal couple probes are inserted into the packed particle at different heights. According to the thermocouple probe at the top side particles, the output temperature value is stable after 3000 s. The structure of the packed particles in the experiment is then replicated by the DEM, which is then input to the LB model for heat conduction simulation. The experimentally measured temperature distributions are compared with the model predicted temperature distribution.

3.2. Comparison between experimental and simulation results

To compare the experimental and simulation results, an air-steel two phases structure is generated for our LBM simulation, in which the steel particle diameter, the height of accumulated packed particle, boundary conditions are the same to the experiment setting. The outline and interface boundary conditions of the LB model are set as given in Table 3.

The simulation results from our LB model and the experimental measurement are compared in Fig. 8, in which the heights mean the

Table 3

Outline and interface boundary condition settings for packed particle heat conduction experiment.

Boundary classification	Surface of sample	Boundary type	Value for setting
Outline boundary	Top surface	Robin BC	Ambient temperature $T_\infty = 291.85 \text{ K}$
	Bottom surface	Dirichlet BC	Bottom temperature $T_{\text{substrate}} = 341.35 \text{ K}$
	Vertical surfaces	Adiabatic BC	
Interface boundary	Air side surfaces at the air/steel interface	Dirichlet BC	T_e at related interface grids of air/steel interface
	Steel side surfaces at the air/steel interface	Dirichlet BC	T_e at related interface grids of air/steel interface

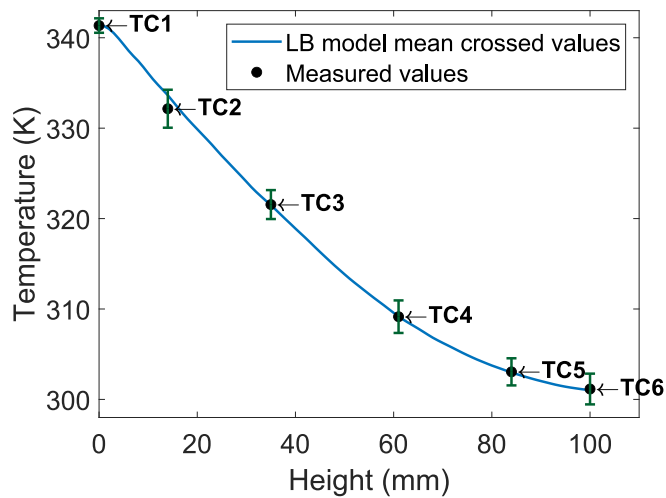


Fig. 8. Comparison of measured temperature with error bars and simulated mean cross temperature in different heights (TC1: 0 (bottom), TC2: 14 mm, TC3: 34.5 mm, TC4: 61 mm, TC5: 84 mm, TC6: 100 mm) for packed particles.

distance between the heated plate and the temperature recorded level, the LB model simulated temperature points are the averaged values of horizontal sections at different heights. It is noted that the experiment was repeated 3 times, and the measured results are taken as the averaged values for enhancing the accuracy and reliability of the measured results. The heights of numbered thermal couple probes for temperature measurement are also depicted in Fig. 8. Those simulation results are obtained after 6×10^5 iterations and the related physical time is 3000 s. Generally, all the simulation results show good consistency with the experimental data, indicating this LB model is suitable for the heat transfer simulation of porous cathodes.

4. Results and discussions

To investigate which parameters could impact on the effective thermal conductivity (ETC), the following parameters: overall porosity, CAM/SE component volume fraction, particle size, tortuosity along the heat conduction direction, and how heterogeneous pore distributions influence the ETCs are examined in this section. The ranges of variation for volume fraction, porosity, and grain particle sizes of both LCO and LLZO for following discussion in Sections 4.1, 4.2, and 4.3 are summarized in Table 4.

4.1. The effect of volume ratio and overall porosity on heat conduction

To explore the effect of LCO/LLZO (CAM/SE) component volume ratios and overall porosities on the heat conduction, 15 different samples are divided into 3 groups as Case I-a/b/c with three different volume ratios, while the minimum and maximum overall porosity are 4.4 % and 31.2 %, respectively. Table 5 summarizes the microstructure parameters in detail. The volume ratio of LCO:LLZO is defined as: Volume Ratio (LCO:LLZO) = Volume of Component LCO/Volume of Component

Table 4

The ranges of variation for volume fraction, porosity, and grain sizes of both LCO and LLZO for composite cathode samples in Sections 4.1, 4.2, and 4.3.

Section	Grain particle sizes (μm)		Volume fraction (LCO: LLZO)	Porosity %
	LCO	LLZO		
4.1	5 μm	5 μm	1:4, 1:1, 4:1	4.4 % to 31.2 %
4.2	5 μm	10 μm	1:1 (fixed)	21.1 % (fixed)
4.3	3 μm to 6 μm	3 μm to 6 μm	3:7 to 7:3	7 % (fixed)

LLZO.

Table 5 presents multiple cases where porosity, component volume ratios and tortuosity values are varied to assess their combined effects on heat conduction, in which “inf” means the tortuosity of related component cannot be measured because the component is not connected across the heat conduction direction. With porosities ranging from 4.4 % to 31.2 %, the tortuosities of pores apparently decline, instead, the tortuosities of LCO and LLZO arise (when the tortuosity can be measured and available), and the ETC decreases significantly (as shown in Fig. 9), demonstrating how higher porosity combined with higher tortuosity of pores deteriorates heat conduction. The results suggest that higher tortuosity of solid-phase and lower tortuosity of pores represent more convoluted and longer paths for heat transfer, effectively increasing thermal resistance.

Fig. 9 shows the 3D temperature distributions of particles for minimum and maximum overall porosities at the same boundary conditions with different component volume fractions (Case I-a1/b1/a5/b5). When the LCO/LLZO volume ratio is 4:1, it can be seen that there is very little temperature difference across the electrode with the lowest overall porosity of 4.4 %. Increasing the overall porosity to 31.2 % causes considerable temperature differences between the inlet (bottom) of 340 K and the outlet (top) of 315 K, which also indicates the reduced thermal conductivity. When the volume ratio drops from 4:1 to 1:1, the uneven temperature distribution becomes more significant, with the inlet temperature at 340 K and the outlet at 295 K. It also means that decreasing the volume fraction of LCO declines the cathode thermal conductivity, as LCO has higher intrinsic thermal conductivity than LLZO. Accordingly, the relationship between temperature gradients across the cathode structure and specific microstructural features is elucidated that higher porosity leads to more significant thermal resistance and uneven heat conduction pathways. Adiabatic vacuum pores disrupt thermal conductivity by introducing localized thermal barriers.

Fig. 9(a) exhibits the temperature distribution within the phases (e.g., LCO and LLZO) and the gradient across the heat conduction direction, which can significantly impact heat conduction in realistic battery systems. The gradient figure is in accordance with the ETC of the related samples in Fig. 10, in which the sample with steeper gradient curve relates to the lower ETC. Fig. 9(b) illustrates temperature gradients within the sole phases, i.e., LCO and LLZO, show the almost same trend and range to the whole solid-phase since the results are in thermal steady-state, but the top side temperature of LLZO for Case I-a5 shows slight difference to LCO and the whole solid-phase. Within high-conductivity phases such as LCO, small temperature gradients can quickly amplify due to its high thermal diffusivity. This can result in localized hot spots that accelerate material degradation or lead to mechanical stresses.

To investigate the relationship between different overall porosities and effective thermal conductivity, cathode structures with five porosities are considered whilst maintaining the same LCO to LLZO volume ratios. Fig. 10 shows the effective thermal conductivity as a function of porosity. The effective thermal conductivity continues to decrease with the increment of the overall porosity for all cases I-a, I-b, I-c. It can confirm that the adiabatic vacuum pores apparently deteriorate heat conduction of the cathode structures. In addition, the volume ratio LCO: LLZO = 1:1 shows higher ETC than the 1:4 ratio case and lower ETC than the 4:1 ratio case, due to the thermal conductivity of LCO is more than ten times that of LLZO. However, the gap between both ratios gradually decreases at higher porosities, indicating the overall porosity has a greater impact on thermal conductivity than volume ratio. The results emphasize the critical role that overall porosity and component volume fraction play in determining the thermal management efficiency of ASSLBs.

To benchmark the LBM simulated ETC values (marked as LBM Simu. k_{eff}) in Fig. 10 by comparing other analytical models, Table 6 compares them with the predicted ETCs by the effective medium theory (EMT)

Table 5

Comparison of three cases of cathode samples with different porosities for Case I-a: “LCO(5 μ m): LLZO(5 μ m) = 1:1”, Case I-b: “LCO(5 μ m): LLZO(5 μ m) = 4:1” and Case I-c: “LCO(5 μ m): LLZO(5 μ m) = 1:4”

Group	Sample	Particle sizes (μ m)		Tortuosity			Volume ratios (LCO: LLZO)	Overall porosity (%)
		LCO	LLZO	Pores	LCO	LLZO		
I-a	I-a1			inf	2.552	2.661	1:1	4.4
	I-a2			inf	2.418	2.839		8.1
	I-a3			50.42	2.824	3.075		15.1
	I-a4			22.30	3.428	3.718		21.1
	I-a5			5.278	4.905	5.436		31.2
I-b	I-b1			inf	1.354	inf	4:1	4.4
	I-b2			inf	1.415	inf		8.1
	I-b3	5	5	50.42	1.596	inf		15.1
	I-b4			22.30	1.801	inf		21.1
	I-b5			5.278	2.319	inf		31.2
I-c	I-c1			inf	inf	1.354	1:4	4.4
	I-c2			inf	inf	1.415		8.1
	I-c3			50.42	inf	1.596		15.1
	I-c4			22.30	inf	1.801		21.1
	I-c5			5.278	inf	2.319		31.2

(denoted as EMT model: k_{eff}). Additionally, it highlights the volume ratios of interconnected solid phases and pores for various cathode microstructures with differing porosities. The EMT assumes a completely random distribution of components, treats the material as a homogenized medium and offers a physically grounded framework for estimating the ETC, which assumes that the components (e.g., active materials and electrolytes) are evenly distributed. This model is particularly significant for porous media, where the complex interplay of solid and void phases dictates heat transfer behaviour. The equation of EMT for a multi-component material is written as Eq. (31), where $d_i = 3$, $\tilde{k} = k_{eff}$, m is the number of components, v is the volume fraction of component [61]. Set Eq. (31) as $k_{eff} = f(k_{eff})$, the ETC can be solved numerically by MATLAB tool “vpasolve” from the symbolic equation $f(k_{eff}) - k_{eff} = 0$.

$$k_{eff} = \frac{\sum_{i=1}^m k_i v_i \frac{\tilde{d}_i \tilde{k}}{(d_i - 1)k + k_i}}{\sum_{i=1}^m v_i \frac{\tilde{d}_i \tilde{k}}{(d_i - 1)k + k_i}} \quad (31)$$

Additionally, for the study of the microstructural features, the analysis of interconnected pores and solid phases impact the thermal conduction pathways in Table 6, in which the volume ratio of interconnected solid phases (LCO and LLZO) remains consistently high across all samples (above 99 %), indicating robust pathways for heat conduction in the solid phase. In contrast, the interconnected pore phase volume ratio increases dramatically with porosity, from 0.213 % in I-a1 to 74.265 % in I-a5. This increase illustrates the dominance of isolated and dispersed voids at lower porosities, transitioning to interconnected pore networks that significantly impair thermal conduction at higher porosities.

The results in Table 6 show that the LBM simulation closely aligns with the EMT predictions for low-porosity samples dominated by high thermal conductivity component LCO (e.g., I-a1 and I-b1), where the ETC is less disrupted by adiabatic vacuum pores. While as porosity increases, deviations between LBM and EMT predictions become pronounced. For high-porosity samples (e.g., I-a5) dominated by high thermal conductivity component LCO, LBM simulations predict significantly lower ETC compared to EMT. However, for samples dominated by low thermal conductivity component LLZO, the low-porosity samples (e.g., I-c1) show the higher deviation (~28 %) than the higher porosity ones (e.g., I-c5), which means the low thermal conductivity component LLZO is a major factor disrupts heat conduction pathways in this condition.

This discrepancy arises because EMT assumes idealized, isotropic material properties and continuous pathways for heat conduction,

failing to account for the more complex and disrupted heat pathways seen in high-porosity structures. The EMT model provides a simplified analytical framework for estimating the ETC of composite materials on Table 6 but does not account for critical microstructural factors such as heterogeneity of component distributions, tortuosity, and particle-scale effects according to Eq. (31). These limitations highlight why EMT, while useful as a benchmark, may fail to accurately predict the ETC of highly heterogeneous and anisotropic systems. Advanced simulation techniques like LBM are essential for capturing these effects, as they explicitly model the microstructural features, including the spatial distribution of components, tortuosity, and particle-scale interactions. For more specific insights into the influence of microstructural features, a group of cathode samples with different tortuosities and similar particle sizes (LCO: 5 μ m, LLZO: 10 μ m), volume ratios (LCO:LLZO = 1:1) and porosities (21.1 %) are simulated in Section 4.2 to investigate the effect of tortuosity on the ETC, where the EMT-assumed ETCs for Case II (i.e., II-a1,2 and II-b1,2,3) are all 1.244 W m⁻¹ K⁻¹ but the simulated ETCs varies from 0.758 W m⁻¹ K⁻¹ to 1.306 W m⁻¹ K⁻¹.

4.2. The effect of tortuosity on heat conduction

Pores within cathode microstructures are non-conductive regions that impede heat flow. Heterogeneous pore distribution introduces anisotropic behaviour in ETC, where the ETC varies along different directions of the cathode. This anisotropy arises from the uneven arrangement of pores, which creates preferential pathways for heat conduction in certain directions while hindering it in others. Understanding how these irregularities in pore distribution affect thermal conductivity is essential for optimizing the design and efficiency of ASSLBs.

Tortuosity is a dimensionless factor that represents the ratio of the actual path length must travel through a medium, compared to the straight-line distance between two points, which influences the pathways for heat conduction. This factor can be used to describe the distribution of components in cathodes, an independent property in each direction of the porous media due to the definition. Understanding and optimizing the tortuosity for heat conduction in the cathode structure is therefore essential. To investigate the effect of tortuosity on heat conduction, two groups of structures, as presented in Table 7 and Fig. 11 with similar tortuosity values (Case II-a), Table 8 and Fig. 12 with different tortuosity values (Case II-b), are generated while the particle sizes, volume ratios and porosities are kept constant. The tortuosity can be calculated for each component (pore, LCO and LLZO) of three dimensions (x-, y-, and z-direction) by TauFactor [35] to discover how heterogeneous pore distribution influences the ETC for ASSLBs. Each composite cathode sample has three ETCs of three dimensions and every

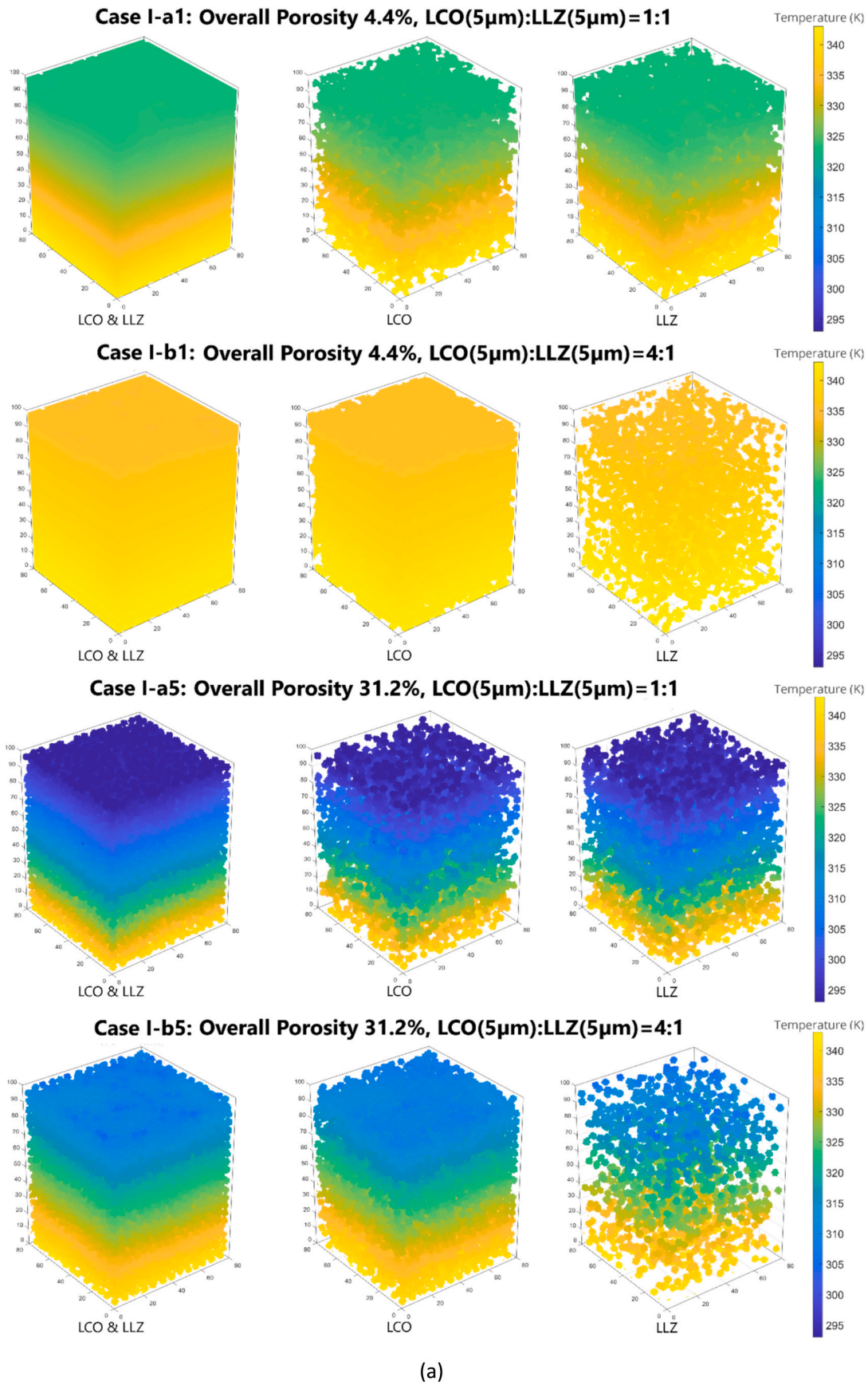


Fig. 9. (a) 3-D temperature distribution graph and (b) gradient across the conduction direction for three phases (i.e., whole solid-phase, LCO and LLZO) of Case I-a1/b1/a5/b5.

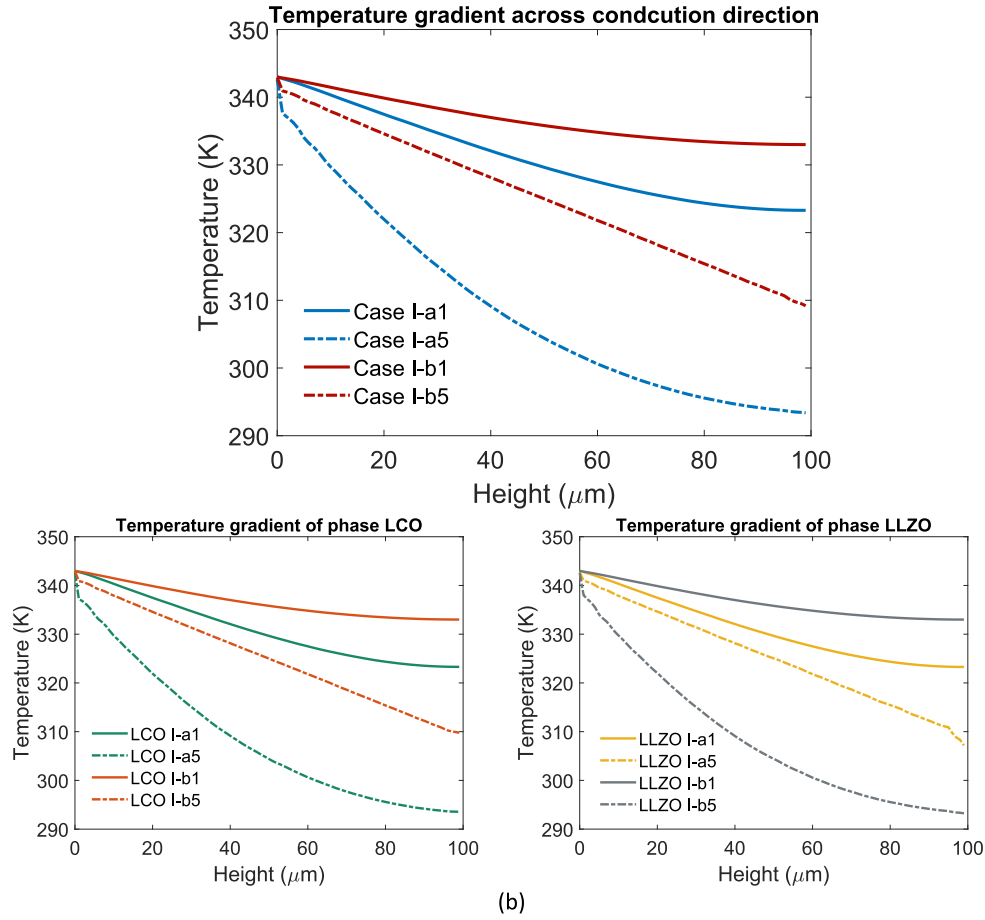


Fig. 9. (continued).

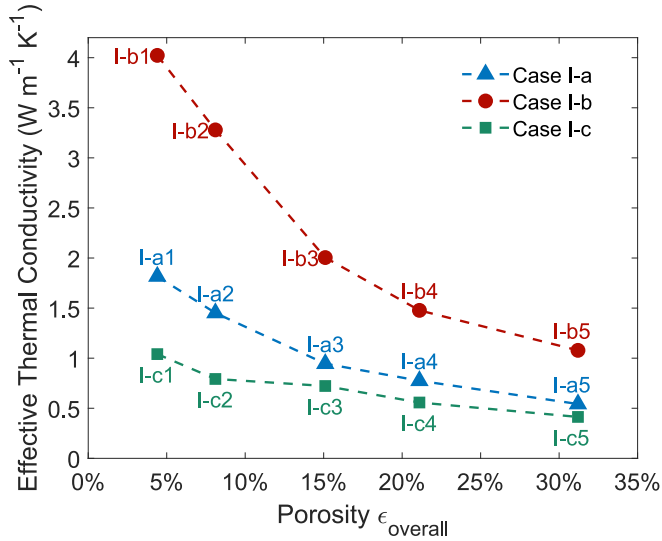


Fig. 10. Comparison of effective thermal conductivity distributions k_{eff} for Case I-a: “LCO(5 μm): LLZO(5 μm) = 1:1”, Case I-b: “LCO(5 μm): LLZO(5 μm) = 4:1” and Case I-c: “LCO(5 μm): LLZO(5 μm) = 1:4”.

ETC is related to three tortuosities of pore, LCO and LLZO.

It should be noted that tortuosity can only be measured properly when the target component has completely connected pathways. If the component fraction is low and is not completely connected along the heat conduction pathways, the tortuosity of this component cannot be

Table 6

Comparison of effective thermal conductivity k_{eff} of LBM simulated and effective medium theory (EMT) model analytical results, volume ratio of interconnected pores and solid material to the related whole phases for Case I-a

Sample	LBM simu. k_{eff} ($\text{W m}^{-1} \text{K}^{-1}$)	EMT model: k_{eff} ($\text{W m}^{-1} \text{K}^{-1}$)	Interconnected Solid phase	Interconnected Pore-phase
I-a1	1.816	1.904	99.965 %	0.213 %
I-a2	1.452	1.753	99.987 %	4.455 %
I-a3	0.945	1.475	99.883 %	11.054 %
I-a4	0.775	1.243	99.871 %	19.897 %
I-a5	0.542	0.874	99.819 %	74.265 %
I-b1	4.023	3.716	99.965 %	0.213 %
I-b2	3.279	3.472	99.987 %	4.455 %
I-b3	2.005	3.014	99.883 %	11.054 %
I-b4	1.477	2.620	99.871 %	19.897 %
I-b5	1.078	1.966	99.819 %	74.265 %
I-c1	1.041	0.751	99.965 %	0.213 %
I-c2	0.792	0.695	99.987 %	4.455 %
I-c3	0.722	0.593	99.883 %	11.054 %
I-c4	0.557	0.509	99.871 %	19.897 %
I-c5	0.413	0.377	99.819 %	74.265 %

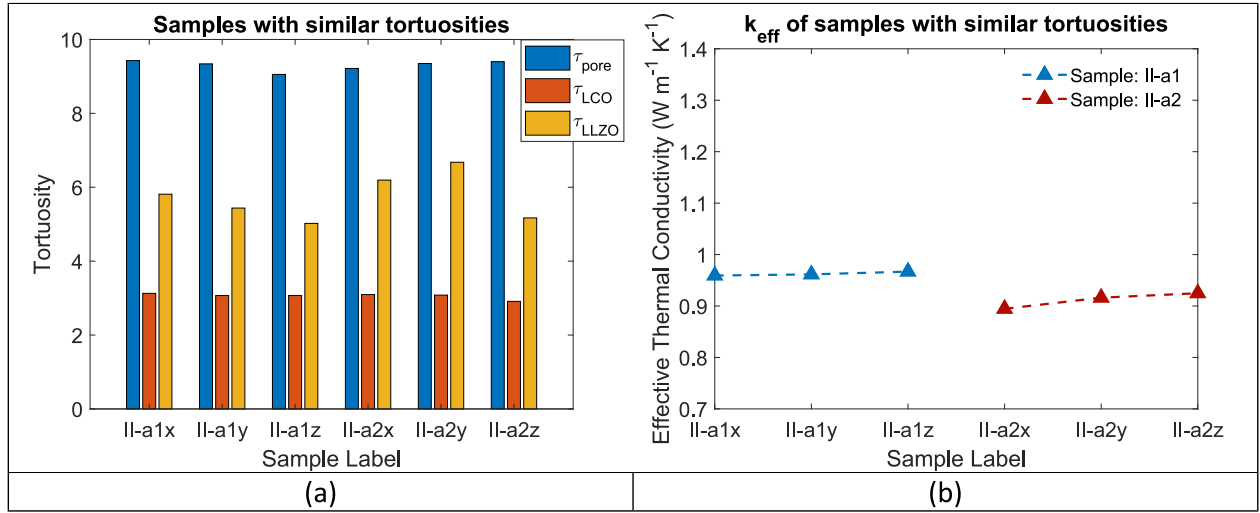
measured. For example, when overall porosity is small (<21.1 %), the tortuosity of pores cannot be fully obtained since some pores are not completely connected along the heat conduction direction. To ensure tortuosity values are obtained, two structures with similar tortuosities as Case II-a in Table 7 and three structures with different tortuosities as Case II-b Table 8 are generated with an overall porosity of 21.1 %. To exclude the effects of other parameters, both Case II-a and Case II-b also maintain a particle size of 5 μm for LCO and 10 μm for LLZO, with a 1:1

Table 7

Case II-a cathode microstructure parameters: first group with similar tortuosity

Sample	Dimension	Particle sizes (μm)		Volume ratios (LCO/LLZO)	Overall porosity (%)	Tortuosity			k_{eff} (W m ⁻¹ K ⁻¹)
		LCO	LLZO			Pores	LCO	LLZO	
II-a1	X	5	10	1:1	21.1	9.428	3.128	5.812	0.959
	Y					9.339	3.069	5.437	0.961
	Z					9.054	3.069	5.022	0.967
II-a2	X	5	10	1:1	21.1	9.215	3.096	6.194	0.895
	Y					9.349	3.081	6.677	0.916
	Z					9.399	2.912	5.17	0.925

Sample	Standard deviation of parameters across the X, Y, and Z dimensions				k_{eff} (W m ⁻¹ K ⁻¹)
	Tortuosity				
	Pores	LCO	LLZO		
II-a1	0.1954	0.0341	0.3952	0.0039	
II-a2	0.0951	0.1022	0.7695	0.0156	
All samples	0.1399	0.0754	0.6354	0.0295	

**Fig. 11.** Sample Case II-a with similar tortuosity across the X, Y, and Z dimensions, overall porosity $\varepsilon_{\text{overall}}=21.1\%$, the particle size: LCO & LLZO are $5\ \mu\text{m}$ and $10\ \mu\text{m}$, respectively, the volume fraction ratio of LCO & LLZO are 1:1: (a) tortuosity for pore, LCO and LLZO, (b) effective thermal conductivity distributions k_{eff} .

volumetric ratio of LCO to LLZO. Tortuosity values of the pores, LCO, and LLZO phases and ETCs are provided for each sample across the X, Y, and Z dimensions, and the standard deviations of these parameters on three dimensions are also presented.

Figs. 11 and 12 complements the data presented in Tables 7 and 8 by providing visual insights into the tortuosity and ETC distributions for the Case II-a and Case II-b samples, respectively. The ETCs of Case II-a are not significantly different in when the tortuosities are similar, which ranges from 0.895 to $0.967\ \text{W m}^{-1} \text{K}^{-1}$ as shown in Table 7 and Fig. 11. Case II-a illustrate the role of microstructural uniformity in promoting consistent ETC across the X, Y, and Z dimensions, whose pores, LCO and LLZO components distributions are homogeneous on all dimensions.

On the contrast to Case II-a, Case II-b presents the ETCs of three samples (II-b1, II-b2, II-b3) are apparently different with varying tortuosity values across the X, Y, and Z dimensions, as shown in Table 8 with a wider range from 0.758 to $1.306\ \text{W m}^{-1} \text{K}^{-1}$. Fig. 12 emphasizes the ETC distributions are varying by the heterogeneity of the same ASSLB cathode microstructure across different dimensions. Case II-b highlight the complex relationship between tortuosity and thermal performance in heterogeneous cathode microstructures.

According to Tables 7 and 8, larger τ_{pore} more like to enhance the ETC by comparing Case II-b2x and II-b2z, whose τ_{pore} and τ_{LCO} are similar. Instead, comparing Case II-a1y and II-b2x with similar τ_{pore} and τ_{LLZO} ,

higher τ_{LCO} looks to reduce the ETC and impede the heat conduction. Additionally, higher τ_{LLZO} more like to decrease the ETC but not obvious by comparing Case II-b1y and II-b2x with similar τ_{pore} and τ_{LCO} .

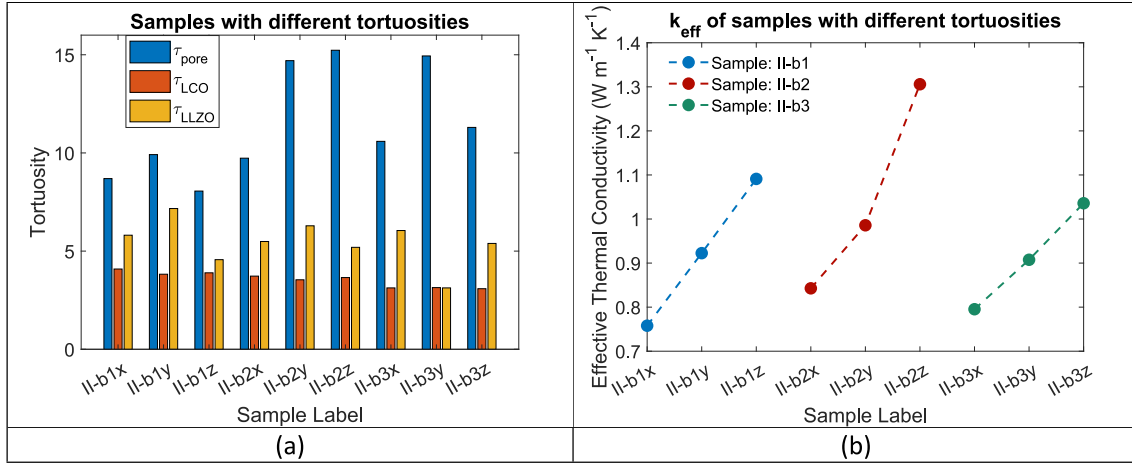
It is strongly consistent with thermodynamics fundamentals: For the tortuosity of the pore phase τ_{pore} , vacuum pore is the strong thermal resistance region and primarily affects the pathways for heat conduction through void spaces in the ASSLB cathode. Higher τ_{pore} means the less connection (higher complexity of the path) of pores and more connection (lower complexity of the path) of solid materials (LCO and LLZO), which implies that the heat conduction through the pores is less direct, potentially allowing more heat to be transferred through the solid materials and increasing the ETC of cathode. LCO is the active material with a significantly higher intrinsic thermal conductivity compared to LLZO or the pore phase. Higher τ_{LCO} increases the complexity and length of the conduction pathways within the LCO phase, which diminishes its effectiveness in facilitating heat transfer. Since LCO is a primary contributor to ETC, increased tortuosity in this phase directly impedes overall heat conduction. LLZO contributes less to overall thermal conductivity compared to LCO due to its lower intrinsic thermal conductivity. Consequently, variations in τ_{LLZO} have a smaller impact on ETC compared to τ_{LCO} . However, increasing τ_{LLZO} still introduces additional thermal resistance, slightly reducing the overall ETC. This observation suggests that while LLZO plays a secondary role in thermal conduction,

Table 8

Case II-b cathode microstructure parameters: second group with different tortuosity

Sample	Dimension	Particle sizes (μm)		Volume ratios (LCO/LLZO)	Overall porosity (%)	Tortuosity			k_{eff} (W m ⁻¹ K ⁻¹)				
		LCO	LLZO			Pores	LCO	LLZO					
II-b1	X	5	10	1:1	21.1	8.693	4.088	5.810	0.758				
	Y					9.915	3.823	7.168	0.923				
	Z					8.058	3.894	4.564	1.091				
II-b2	X					9.733	3.726	5.49	0.843				
	Y					14.70	3.539	6.289	0.986				
	Z					15.23	3.653	5.192	1.306				
II-b3	X									10.59	3.126	6.048	0.795
	Y									14.94	3.141	3.126	0.908
	Z									11.30	3.085	5.391	1.036

Sample	Standard deviation of parameters across the X, Y, and Z dimensions			
	Tortuosity			k_{eff} (W m ⁻¹ K ⁻¹)
	Pores	LCO	LLZO	
II-b1	0.9438	0.1372	1.3024	0.1666
II-b2	3.0323	0.0942	0.5672	0.2371
II-b3	2.3337	0.0290	1.5330	0.1203
All samples	2.7884	0.3684	1.1402	0.1692

**Fig. 12.** Sample Case II-b with different tortuosity across the X, Y, and Z dimensions, overall porosity $\varepsilon_{\text{overall}}=21.1\%$, the particle size: LCO & LLZO are $5\ \mu\text{m}$ and $10\ \mu\text{m}$, respectively, the volume fraction ratio of LCO & LLZO are 1:1: (a) tortuosity for pore, LCO and LLZO, (b) effective thermal conductivity distributions k_{eff} .

its microstructural optimization remains important, especially in systems where it forms a significant fraction of the composite.

The major gap of ETCs is contributed by the tortuosity three components (pore, LCO and LLZO). Using multivariate polynomial regression (MPR) to relate the dependent variable (ETC) to independent variables (tortuosity of pores, LCO, and LLZO) as discussed in the document is an excellent approach. The regression helps to distinguish if they have strong relationship and capture interactive effects of microstructural properties on heat conduction.

By referring to the regression methodology in a previously published paper [62], the p th-order MPR formula relates the response (dependent) variable (y) to explanatory (independent) variables (x) [63] can be written as:

$$y = b_0 + \sum_{i_1=1}^n b_{i_1} x_{i_1} + \sum_{i_1=1}^n \sum_{i_2=1}^n b_{i_1 i_2} x_{i_1} x_{i_2} + \sum_{i_1=1}^n \sum_{i_2=1}^n \sum_{i_3=1}^n b_{i_1 i_2 i_3} x_{i_1} x_{i_2} x_{i_3} + \dots + \sum_{i_1=1}^n \sum_{i_2=1}^n \dots \sum_{i_p=1}^n b_{i_1 i_2 \dots i_p} x_{i_1} x_{i_2} \dots x_{i_p} + \epsilon \quad (32)$$

where $x_1, x_2, \dots, x_n$ represent the input variables of the system, $b_{i_1}, b_{i_1 i_2}, \dots$

, $b_{i_1 i_2 \dots i_p}$ are regression coefficients with p th-order. ϵ is the model's error term (also known as the residuals), which provides for random variation in y not explained by the x variables. The second term in Eq. (32) represents the linear term, the third term represents the quadratic terms, and so on, up to the p th term representing the p th-order interaction terms.

Standard deviations for tortuosity confirm these differences, highlighting the range of configurations used to explore the effects of anisotropic pathways on ETC. By comparing the tortuosities of Case II-a with similar ETCs by Table 7 and Fig. 11, the standard deviation of LLZO tortuosity is the largest, which means the larger change of τ_{LLZO} lead to slight effect on ETC for Case II-a. And the ratio of the standard deviation for all samples of Case II-b to II-a across all dimensions are 19.9, 4.9, 1.8 times for the tortuosities of pore, LCO and LLZO ($\tau_{\text{pore}}, \tau_{\text{LCO}}, \tau_{\text{LLZO}}$), and 5.7 times for the ETC according to Tables 7 and 8, respectively. The standard deviation for τ_{LLZO} is the smallest one and much smaller than the ETC by comparing Case II-a and II-b. Considering the values of $\tau_{\text{pore}}, \tau_{\text{LCO}}, \tau_{\text{LLZO}}$ are not independent with each other, instead, they are coupled and influenced by one another for heat conduction, τ_{LLZO} may have complex relation to other tortuosities and the ETC.

The ETC-tortuosity ($k_{eff} - \tau$) relationship of Case II-a and II-b are used for fitting the MPR formula, considering the component amount is three, the order of MPR is set as $p = 3$. Due to the guess and discussion above that τ_{LLZO} may have complex relation to other tortuosities and the ETC, the highest order of τ_{LLZO} is set as 3, and the highest order of τ_{pore} and τ_{LCO} are set as 1. Therefore, the $k_{eff} - \tau$ MPR formula is trained by the data in Table 7 and Table 8, and it can be written as:

$$k_{eff} = 590.41\tau_{LLZO} - 125.64\tau_{LLZO}^2 + 8.6998\tau_{LLZO}^3 - 44.4\tau_{pore}\tau_{LLZO} + 12.302\tau_{LCO}\tau_{LLZO} - 3.0423\tau_{LCO}\tau_{LLZO}^2 + 5.9615\tau_{pore}\tau_{LLZO}^2 + 0.83769\tau_{pore}\tau_{LCO}\tau_{LLZO} - 8.1223\tau_{LCO} + 80.809\tau_{pore} - 2.6834\tau_{pore}\tau_{LCO} - 885.15 \quad (33)$$

Eq. (33) is calculated by using the MATLAB script *MultiPolyRegress*, which is written by Ahmet Cecen and has been applied in the published paper [64]. To assess whether the MPR predicted k_{eff} have strong relationship, the scatters of the MPR predicted ETC values k_{eff} are plotted and compared with $R^2 = 1$ fitting line (perfect linear relationship between the MPR predicted results), as shown in Fig. 13. The coefficient of determination R^2 and the probability value (p -value) are the common indicators used to estimate the error variance and fitting degree. Generally, when R^2 is close to 1, and the p -value $P \leq 0.05$, it can be determined that a significant regression relationship exists between the response variable y and explanatory variables x . The R^2 statistic and p -value for Eq. (33) are 0.9852 and 2.8119×10^{-13} , respectively, confirming that there is an obvious 3rd-order MPR relationship for the ETC k_{eff} and three tortuosity values τ_{pore} , τ_{LCO} , τ_{LLZO} .

Eq. (33) incorporates terms up to the third order for LLZO tortuosity τ_{LLZO} , while maintaining a first-order relationship for τ_{pore} , τ_{LCO} , and highlight the interdependencies among the three tortuosities τ_{pore} , τ_{LCO} , τ_{LLZO} as their multiply terms. It fully exhibits the complex and non-linear relationship between tortuosities of pores, active material (LCO), solid electrolyte (LLZO) and ETC.

While the DEM method generates cathode samples randomly, it is noted that the tortuosities of most samples are clustered around similar values, as shown in Table 7. This reflects the inherent statistical consistency of the DEM process, where randomly packed particles tend to form microstructures with predictable and comparable tortuosity distributions due to physical constraints in packing. The similar tortuosity values observed in these samples provide a reliable baseline for understanding the relationship between microstructural properties and ETC under average conditions. The samples in Table 8, by contrast, are selected from a larger pool of generated cathode structures, specifically targeting those with tortuosities that deviate significantly from the

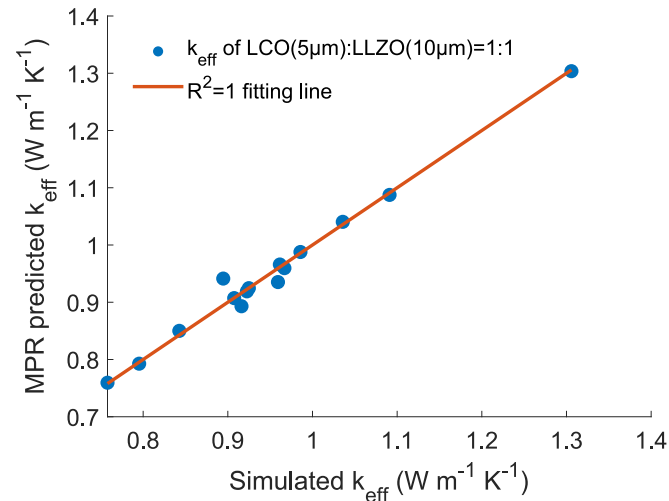


Fig. 13. The MPR predicted - Simulated effective thermal conductivity k_{eff} graph for the cathodes, overall porosity of samples is $\epsilon_{overall}=21.1\%$.

typical range. This deliberate selection provides an opportunity to investigate edge cases and the boundaries of the relationship between tortuosity and the ETC.

The findings underscore the significance of controlling tortuosity to optimize heat dissipation and improve the overall thermal management of ASSLB cathodes. It is worth noting that electrodes with the same average properties of overall porosity, volume ratio and particle size could show very different internal structures and 3D spatial distributions of particles.

In conclusion, heterogeneous distributions of pore, CAM(LCO), SE (LLZO) may result in anisotropic thermal behaviour, where thermal conductivity varies significantly along different directions of the cathode, complicating thermal management strategies. Moreover, not only overall porosity and component volumetric fraction need to be approximate, but the tortuosities of all species should also be similar to keep the cathode REV samples consistent to other samples and the whole cathode.

4.3. The effect of particle size on heat conduction

Particle size is a crucial microstructural parameter in the design of ASSLB cathodes. Previous research, e.g., Al-Jaljouli, Fadi, et al. [65], investigated the relation between the effective electronic and ionic conductivities and grain sizes of LCO and LLZO. However, the effects of CAM(LCO) and SE(LLZO) particle sizes on heat conduction for composite cathodes were barely discovered in previous studies. To explore the impact of particle sizes of LCO and LLZO on heat conduction of cathodes, this section simulates cathode samples with different LCO and LLZO particle sizes. The heat conduction analysis is divided into five groups as Case III-a to III-e and five samples are analysed for each group, where the data is summarized in Table 9 and visualized in Fig. 14, providing insights into the role of particle size in determining the ETC of composite cathodes in ASSLBs. In Table 9, the overall porosity is kept as constant as 7 % for all samples, each group evaluates the ETC across varying LCO volumetric fractions of solid materials from 30 % to 70 % (named as CAM vol% on the table). For particle size configurations, Case III-a has equal particle sizes of 3 μm for LCO (cathode active material, CAM) and LLZO (solid electrolyte, SE), Case III-b shows larger LCO particles of 6 μm relative to LLZO of 3 μm Case III-c exhibit smaller LCO particles of 3 μm than LLZO as 6 μm , and Case III-d exhibits uniformly larger 6 μm particles for both LCO and LLZO. Case III-e uses an intermediate particle size of 5 μm for both LCO and LLZO, compared to Cases III-a and III-d. It is noticed that cathode samples with different particle sizes usually have varying tortuosities when the overall porosity and component volumetric fractions are approximate.

Fig. 14 complements Table 9 by visualizing the ETC trends across the three particle size configurations. The graph illustrates how ETC changes with increasing LCO volumetric fraction for each case. This arrangement allows for a systematic evaluation of how particle size variations influence heat conduction under constant porosity and certain volume fraction.

According to Table 9 and Fig. 14, the ETC generally increases with a higher volumetric fraction of LCO, irrespective of particle size configurations. This trend of higher volume fraction of LCO (the more thermal conductive phase) leading to higher ETC of the composite cathode agrees well with the conclusion in Section 4.1. What is interesting here is that, for all the five cases, there is a clear decrease of the tortuosity of the LCO phase and an increase of the tortuosity of the LLZO phase, when the volume fraction of LCO increases. These results confirm that a higher volume fraction of LCO creates more connected and straighter heat conduction pathways, consequently increasing the ETC of the composite cathodes.

Case III-a (small equal particle sizes of 3 μm for both LCO and LLZO) exhibited better ETC values across all volumetric fractions compared to Case III-b (larger LCO particles of 6 μm), Case III-c (larger LLZO particles of 6 μm), Case III-d (large equal particle sizes of 6 μm for both LCO and

Table 9

Case III: Comparison of three groups of cathode samples with varying particle sizes and overall porosity $\epsilon_{\text{overall}} = 7\%$ and different LCO & LLZO diameters for Case III-a: “LCO:LLZO = 3 μm :3 μm ”, Case III-b: “LCO:LLZO = 6 μm :3 μm ”, Case III-c: “LCO:LLZO = 3 μm :6 μm ”, Case III-d: “LCO:LLZO = 6 μm :6 μm ”, and Case III-e: “LCO:LLZO = 5 μm :5 μm ”.

Group	Sample	Particle sizes (μm)		CAM vol%: LCO/(LCO+LLZO)	Volume ratios (LCO/LLZO)	Tortuosity		Overall porosity (%)	k_{eff} ($\text{W m}^{-1} \text{K}^{-1}$)
		LCO	LLZO			LCO	LLZO		
Case III-a	III-a1	3	3	30 %	3:7	13.22	1.767	7	1.280
	III-a2			40 %	4:6	4.900	2.232		1.689
	III-a3			50 %	5:5	3.046	3.103		2.113
	III-a4			60 %	6:4	2.179	5.059		2.978
	III-a5			70 %	7:3	1.740	14.80		4.005
Case III-b	III-b1	6	3	30 %	3:7	31.88	1.568		1.148
	III-b2			40 %	4:6	5.691	1.847		1.419
	III-b3			50 %	5:5	3.103	2.334		1.770
	III-b4			60 %	6:4	2.167	3.240		2.312
	III-b5			70 %	7:3	1.705	6.142		2.924
Case III-c	III-c1	3	6	30 %	3:7	6.177	1.714		1.203
	III-c2			40 %	4:6	3.270	2.179		1.421
	III-c3			50 %	5:5	2.339	3.125		1.729
	III-c4			60 %	6:4	1.850	5.737		2.139
	III-c5			70 %	7:3	1.569	32.02		2.753
Case III-d	III-d1	6	6	30 %	3:7	15.24	1.623		1.068
	III-d2			40 %	4:6	4.712	1.983		1.174
	III-d3			50 %	5:5	2.707	2.572		1.491
	III-d4			60 %	6:4	2.069	4.211		1.855
	III-d5			70 %	7:3	1.614	11.82		2.505
Case III-e	III-e1	5	5	30 %	3:7	11.91	1.661		1.095
	III-e2			40 %	4:6	4.345	2.076		1.236
	III-e3			50 %	5:5	2.761	2.893		1.561
	III-e4			60 %	6:4	1.994	4.577		1.942
	III-e5			70 %	7:3	1.619	12.99		2.608

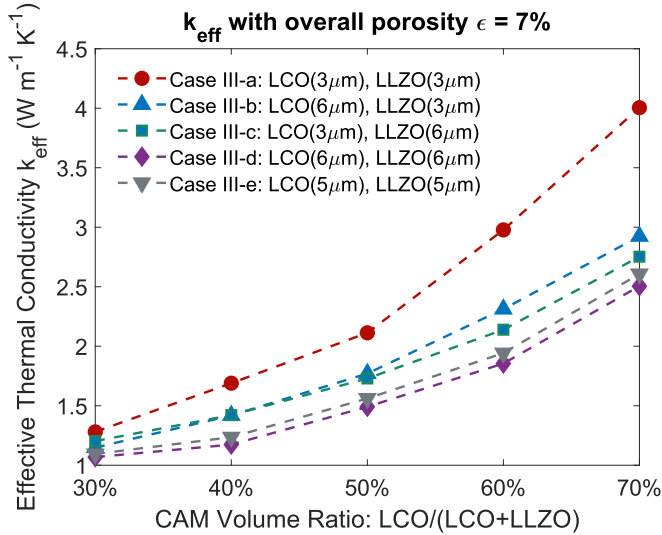


Fig. 14. Comparison of three groups of cathode samples with varying particle sizes and overall porosity $\epsilon_{\text{overall}} = 7\%$ and different LCO & LLZO diameters.

LLZO) and Case III-e (intermediate equal particle sizes of 5 μm for both LCO and LLZO) across all LCO volumetric fractions (30 % to 70 %). Case III-d exhibits the lowest ETC, emphasizing the impact of larger particles on reducing interfacial contact and conduction efficiency. On the other hand, Case III-e serves as an intermediate particle size compared to III-a and III-d, and the ETC of III-e are higher than III-d but obviously lower than III-a. These results indicate that smaller LCO and LLZO particles facilitate better thermal pathways. This is because at the same volume fraction, there are more smaller particles to provide enhanced connected heat conduction pathways, hence giving higher ETC values.

In summary, the particle size plays an important role in determining the ETC and heat conduction in ASSLB cathodes, with smaller particles giving better heat conduction and higher ETC values. The composite

cathodes with larger particle sizes generally has lower thermal conductivity, creating more convoluted pathways for heat conduction and increases the thermal resistance.

5. Conclusions

A 3D lattice Boltzmann heat conduction model was developed to investigate the influence of composite cathode microstructural properties on thermal conduction in ASSLBs. The composite cathode microstructures were generated using DEM simulations with varying structural parameters including volume ratio between cathode active material (CAM) and solid electrolyte (SE) (e.g., LCO and LLZO), overall porosity, tortuosity, and particle size. The effective thermal conductivities of various cathode microstructures were analysed in relation to the microstructural properties. The approach we present here can be extended to investigate other composite cathodes in ASSLBs or composite cathodes in all-solid-state sodium batteries.

The key findings of the current investigation are summarized below:

- The volumetric fraction of LCO and LLZO and the overall porosity are two dominant factors influencing the ETC. Increasing the volumetric ratio of high-conductivity materials, such as LCO, significantly enhances heat conduction and the effective thermal conductivity (ETC). And higher porosity introduces more thermal resistance, leading to a pronounced decrease in ETC. When the overall porosity is low, a high volumetric ratio of LCO can significantly increase the heat conduction performance. Instead, when the overall porosity is high, the adiabatic vacuum pores apparently deteriorate heat conduction. This underscores the critical role of phase composition in optimizing thermal pathways within the cathode structure.
- The effect of tortuosity along the heat conduction direction cannot be ignored on heat conduction. Increasing the tortuosity of pores and LCO has positive impact on thermal conductivity, while reducing the LLZO tortuosity can increase thermal conductivity. But compared to the influence of component volume fraction and overall porosity on ETC, the tortuosity less affects the heat conduction. To make sure the simulated cathode sample can be deemed as the representative

elementary volume (REV), not only overall porosity and component volumetric fraction need to be approximate to the whole cathode, but the tortuosities of all species between the sample and the whole cathode should also be similar.

- Particle size plays an important role in heat conduction. With the same volume fraction, smaller particles create more connected heat conduction pathways and decrease the tortuosity of the solid phase, hence enhance heat transfer.

Our findings demonstrate that optimizing the cathode microstructure to enhance thermal conductivity directly reduces thermal resistance and supports uniform temperature distribution, critical for prolonging ASSLB cycle life and operational reliability. While this study provides foundational insights into the thermal behaviour of pristine composite cathodes, future research should aim to extend the model to include real-world factors that could influence thermal pathways by altering particle contacts and interfacial areas, e.g., particle cracking, loss of contact between active materials and solid electrolytes, irregular particle shapes, and the formation of micro voids during cycling. Incorporating transient heat transfer simulations can further enhance the model's capability to predict temperature variations under dynamic operating conditions. And a formal sensitivity analysis to evaluate the robustness of the results to variations in porosity, particle size, tortuosity, and volume ratio would identify critical parameters and provide deeper insights into the reliability of ETC predictions. These developments will allow for a more accurate representation of long-term performance and reliability in practical applications.

Nomenclature

c_k	the lattice speed
c_s	the speed of sound (m/s)
c_p	specific heat capacity (kJ kg ⁻¹ K ⁻¹)
e	discrete lattice velocity
$\mathcal{F}$	external force term
f	lattice Boltzmann distribution function for fluid field
g	lattice Boltzmann distribution function for diffusivity
$\dot{q}$	heat generation term
l	length (m)
k	thermal conductivity (W m ⁻¹ K ⁻¹)
h	average convective heat transfer coefficient (W m ⁻² K ⁻¹)
q	heat flux (W m ⁻²)
S	heat source term (K s ⁻¹)
T	temperature (K)
t	time (s)
v	volume fraction of component
w	weight coefficient

Greek symbols

α_T	thermal diffusivity (m ² s ⁻¹)
Δ	the increment of a quantity
Δt	time step
Δx	mesh scale
ρ	density / mass concentration (kg m ⁻³)
τ_T	relaxation time
ω	relaxation coefficient
Ω	collision term

Subscripts

α	lattice direction
i	component i

Superscripts

eq equilibrium

CRediT authorship contribution statement

Juan Huang: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Jiawei Hu:** Software, Data curation. **Duo Zhang:** Methodology. **Yuheng Du:** Writing – review & editing, Validation, Formal analysis. **Chuan-Yu Wu:** Writing – review & editing, Supervision. **Qiong Cai:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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